



Autumn 2025

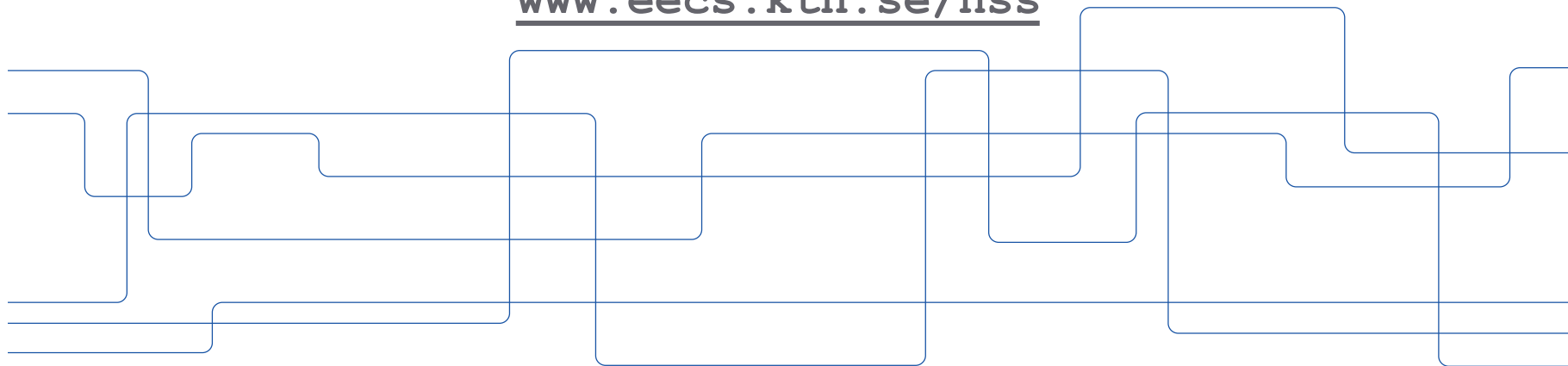
Secure Routing

Ad hoc networks and BGP

Discussion leads
Marco and Ahmed

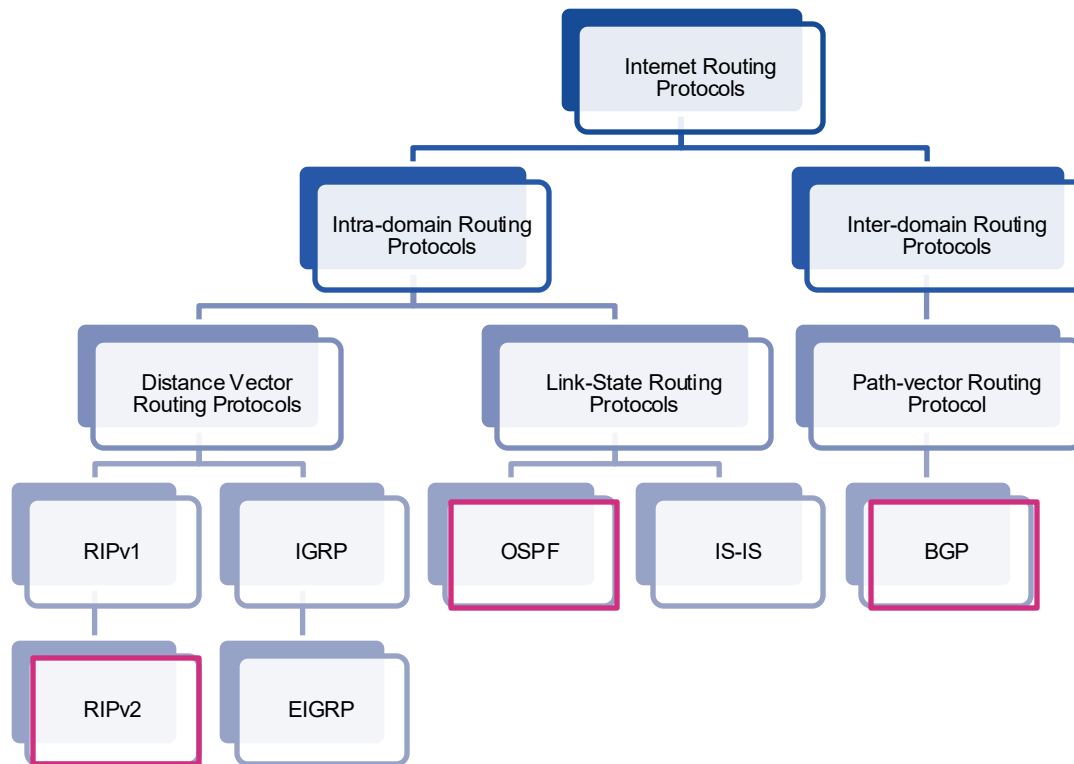
Panos Papadimitratos

www.eecs.kth.se/nss



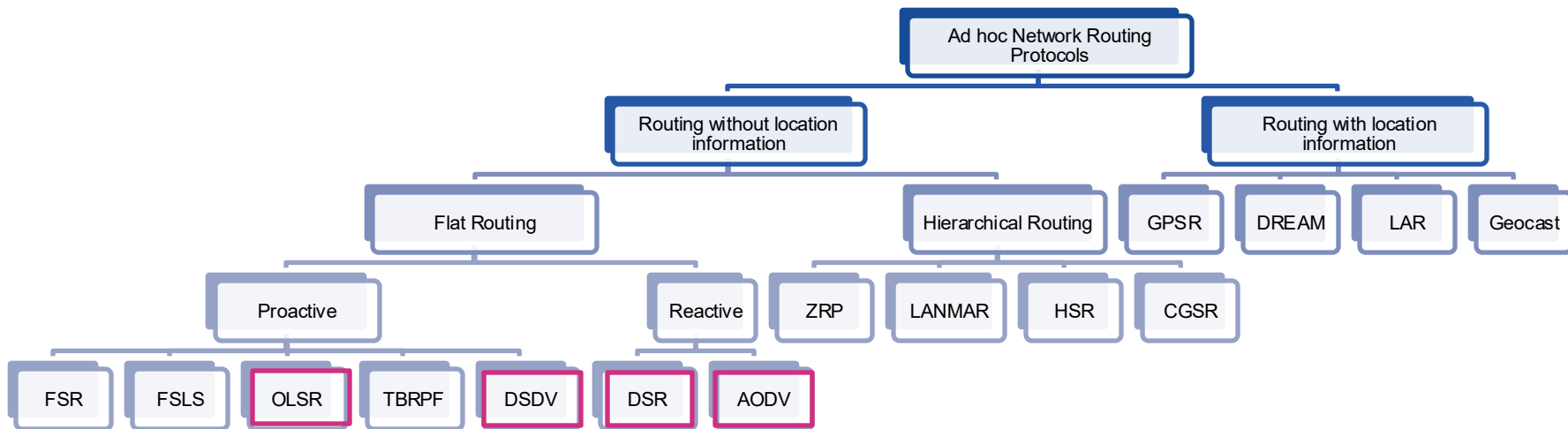
Introduction

Internet Routing Protocols



Introduction (cont'd)

Ad hoc Network Routing Protocols





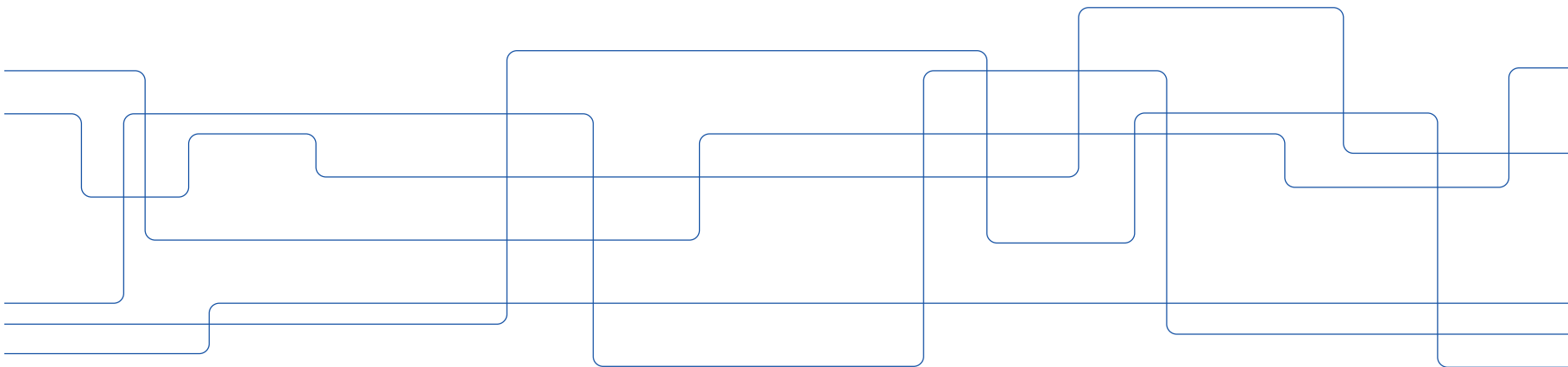
Reading list

1. W. Galuba, P. Papadimitratos, M. Poturalski, K. Aberer, Z. Despotovic and W. Kellerer, "Castor: Scalable Secure Routing for Ad Hoc Networks," 2010 Proceedings IEEE INFOCOM, 2010, pp. 1-9, doi: 10.1109/INFCOM.2010.5462115.
2. K. Shrishak and H. Shulman, "Privacy Preserving and Resilient RPKI," IEEE INFOCOM 2021 - IEEE Conference on Computer Communications, 2021, pp. 1-10, doi: 10.1109/INFOCOM42981.2021.9488759.

Secure Route Discovery

Ad hoc networks

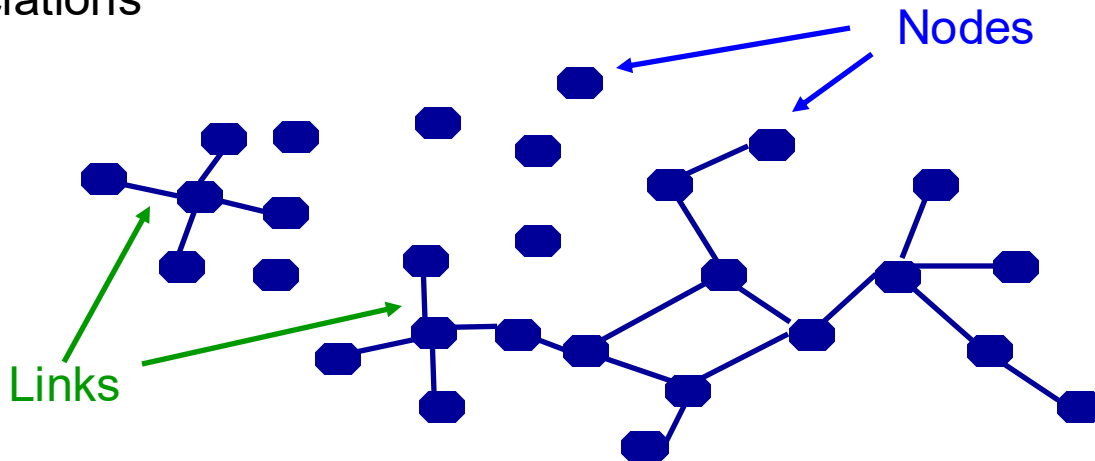
SRP, SLSP, CASTOR



Introduction

Wireless Systems

- Ad hoc networks
 - Limited wireless communication range
 - Collaborative support of the network operation
 - Peer-to-peer interactions
 - Transient associations
 - Openness





Introduction (cont'd)

Wireless Systems

- Security challenges
 - Easy eavesdropping and message injection
 - Each and every node can disrupt the network operation
 - No monitoring facility
 - Resource constraints
 - Error-prone communication
 - Hostile environments
 - Nodes and applications tightly coupled to the user and her physical environment



Introduction (cont'd)

Wireless Systems

A set of basic elements

Radio link establishment

Direct wireless
communication

Distance to other reachable
devices

Multi-hop communication

Device localization and
own positioning

Application performance measurable in
the physical world



Introduction (cont'd)

Wireless Systems

A set of basic elements

Anti-jamming techniques

Secure Neighbor
Discovery

Secure ranging
Distance bounding

Secure data
communication

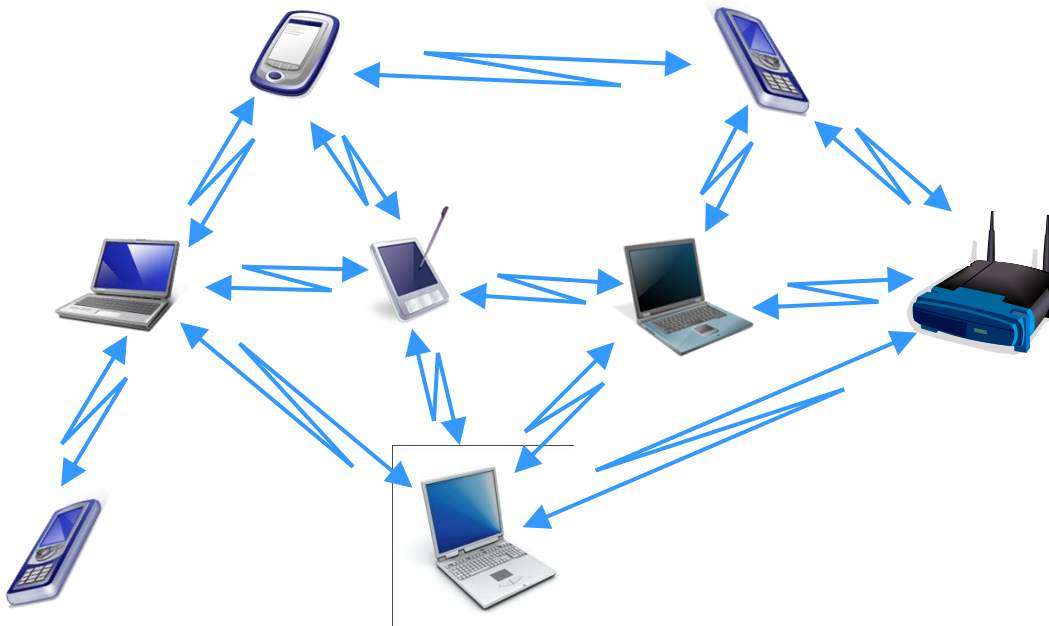
Secure localization and
positioning

Secure Vehicular Communications –
transportation safety

Introduction (cont'd)

Wireless Systems

- Wireless multi-hop connectivity

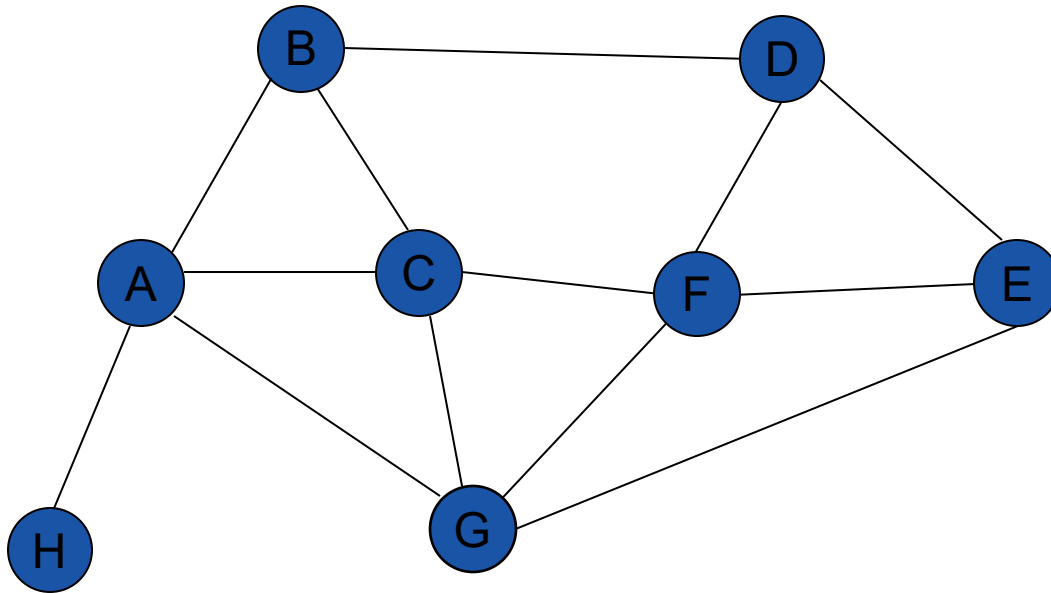




Introduction (cont'd)

Wireless Systems

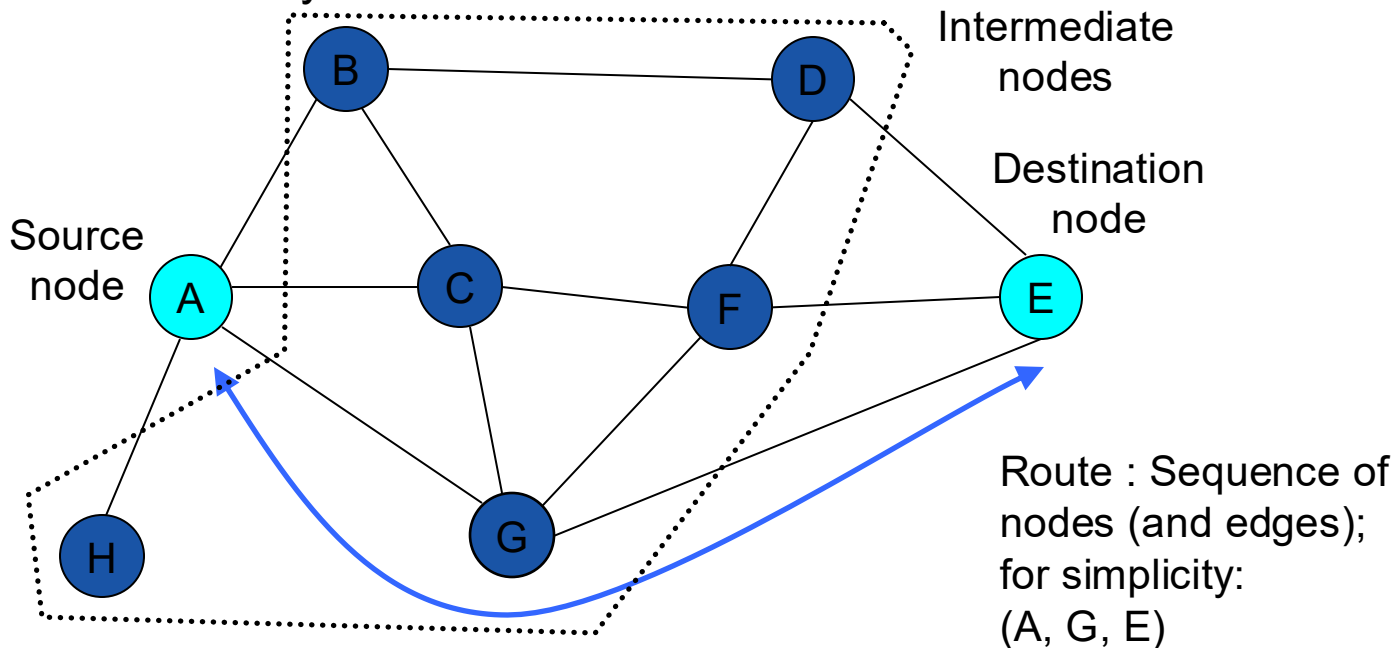
- (Multi-hop) Connectivity graph



Introduction (cont'd)

Route discovery

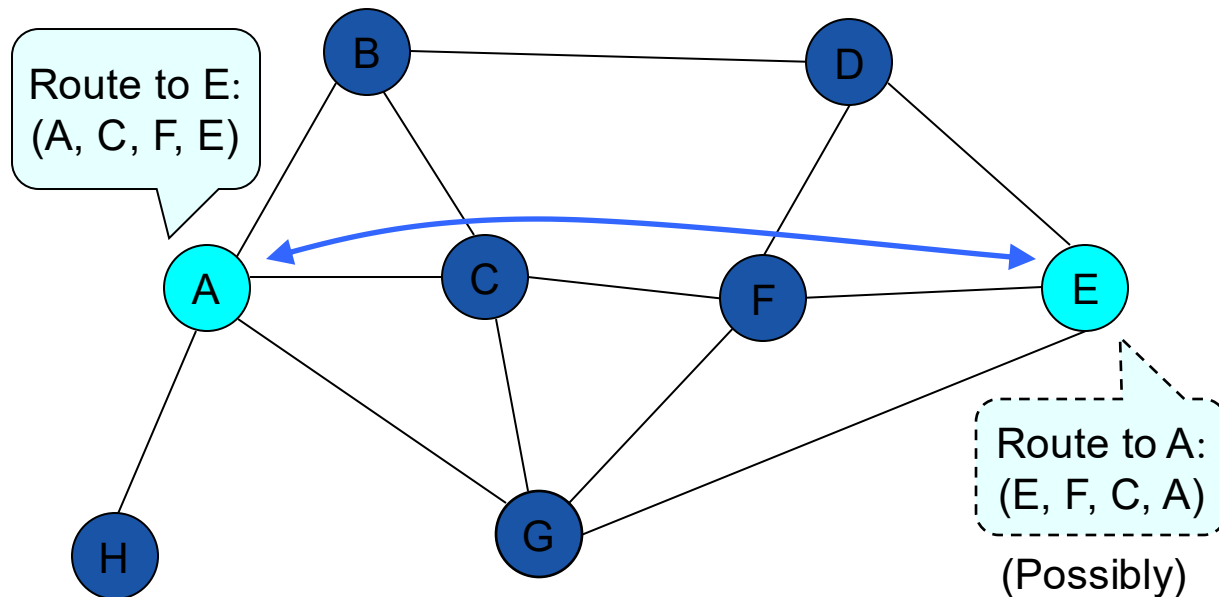
- Stage 0: neighbor discovery
- Stage 1: route discovery



Introduction (cont'd)

Route discovery

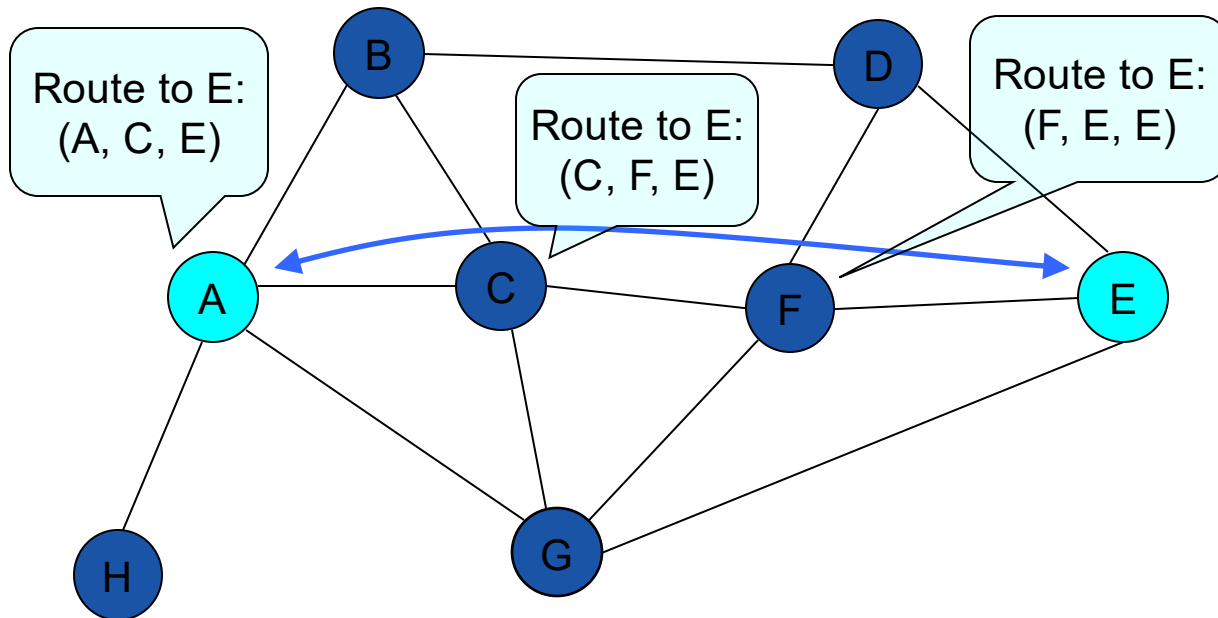
- Explicit route discovery
 - Fully, clearly expressed and readily observable route returned by the routing protocol



Introduction (cont'd)

Route discovery

- Implicit route discovery
 - Distributed computation that returns a tuple of the form (current node, relay node, destination node)





Introduction (cont'd)

Route discovery

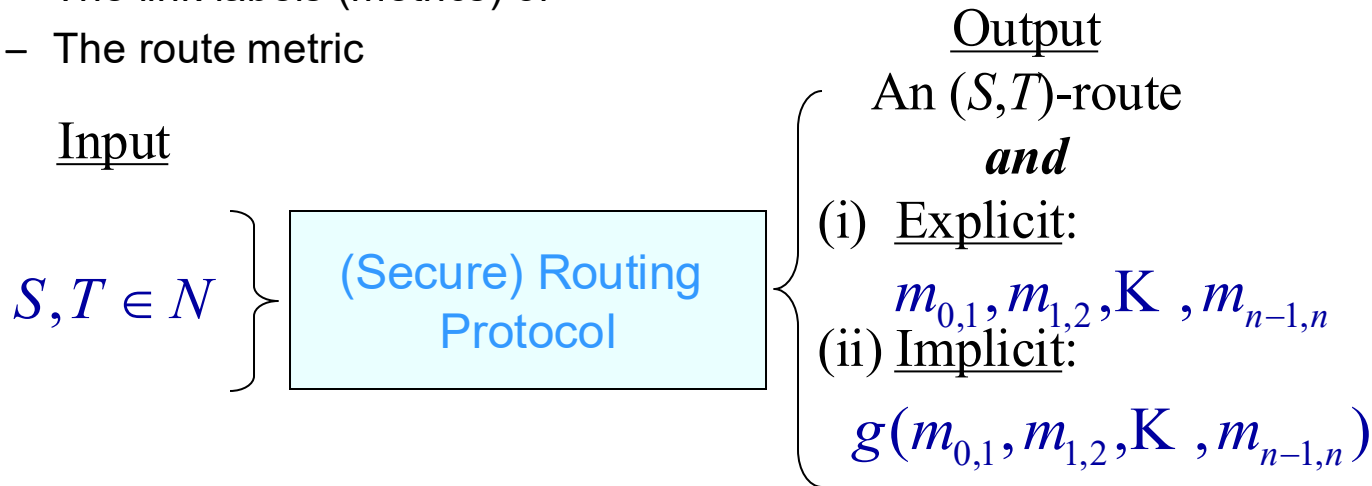
- Basic route discovery
 - Explicit or implicit, providing only the structure of the route
- Augmented route discovery
 - Need a function that assigns labels to links, denoted as link metrics
 - > For a link (V_1, V_2) , metric $m_{1,2}$
 - Route metric: a function that is the aggregate of the route link metrics
 - > For a route $(V_0, V_1, \dots, V_n)$, route metric
$$g(m_{0,1}, m_{1,2}, \dots, m_{n-1,n})$$



Introduction (cont'd)

Route discovery

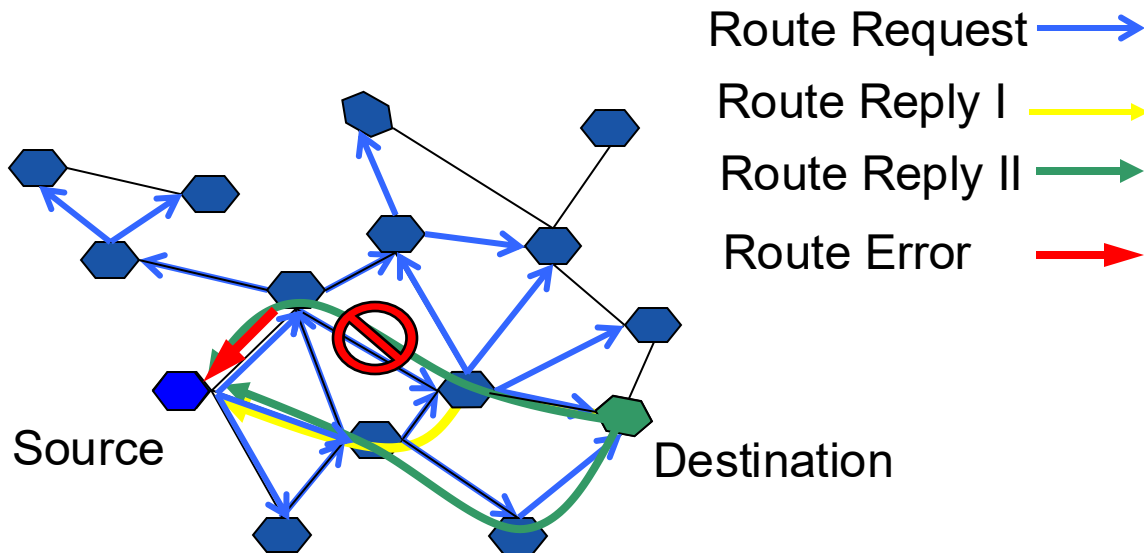
- Input: source, S , and destination, T , nodes
- Output: an $(S-T)$ -route (of n links) and
 - The link labels (metrics) or
 - The route metric



Introduction (cont'd)

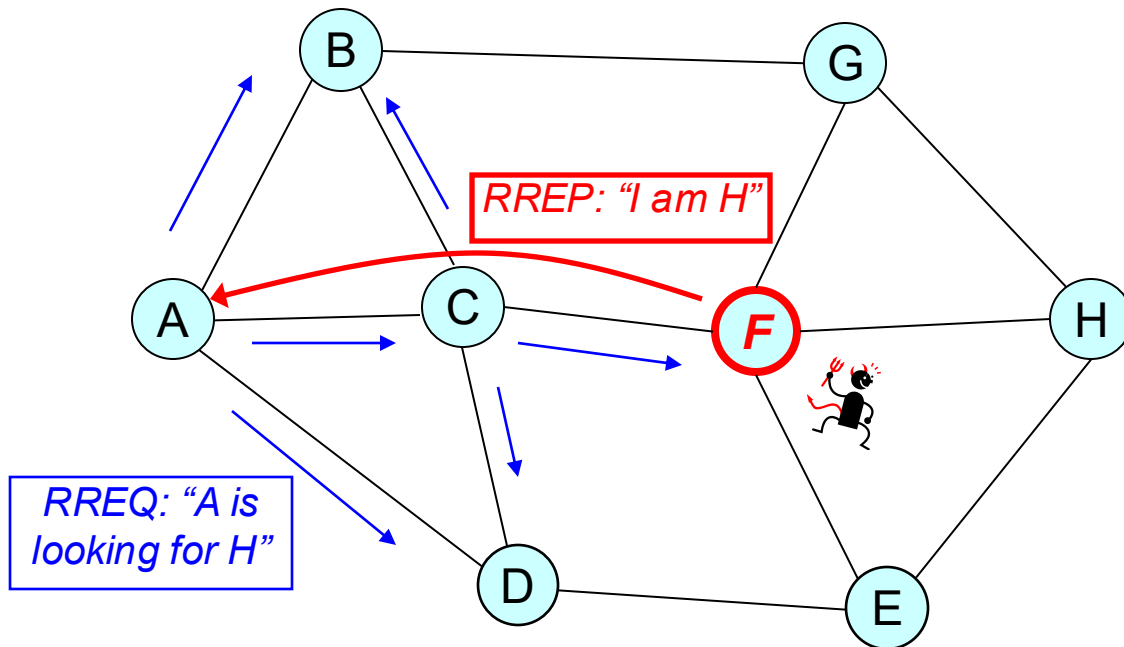
Route discovery

- Example of route discovery
 - Reactive routing protocol



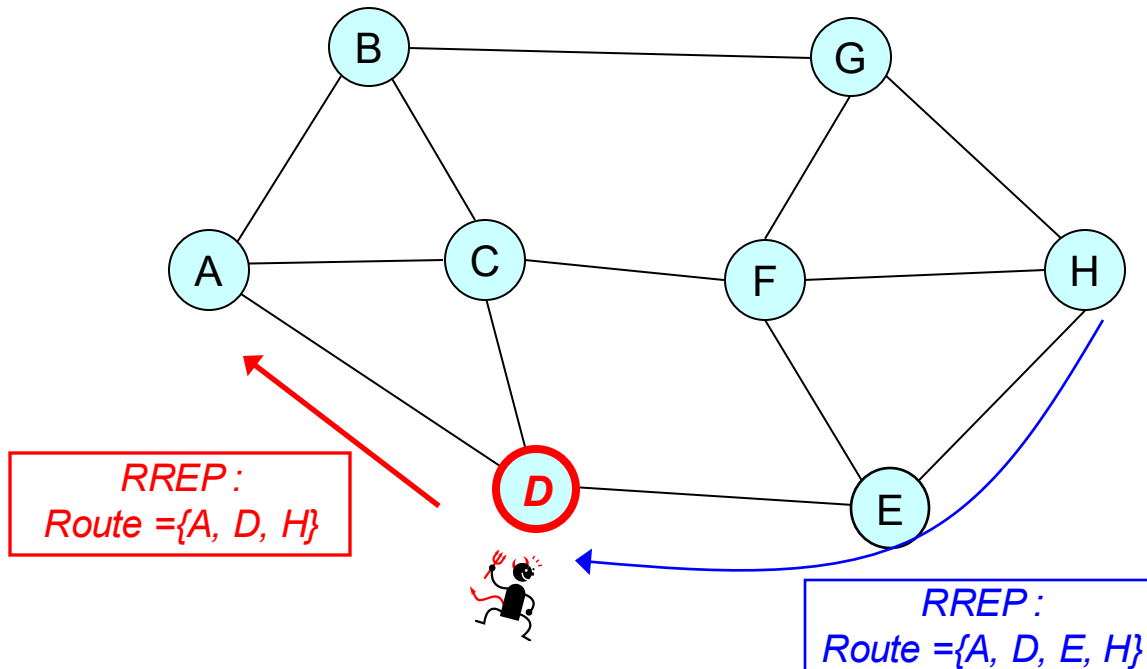
Attacking route discovery

- Impersonation of the destination, for example, in any reactive routing protocol



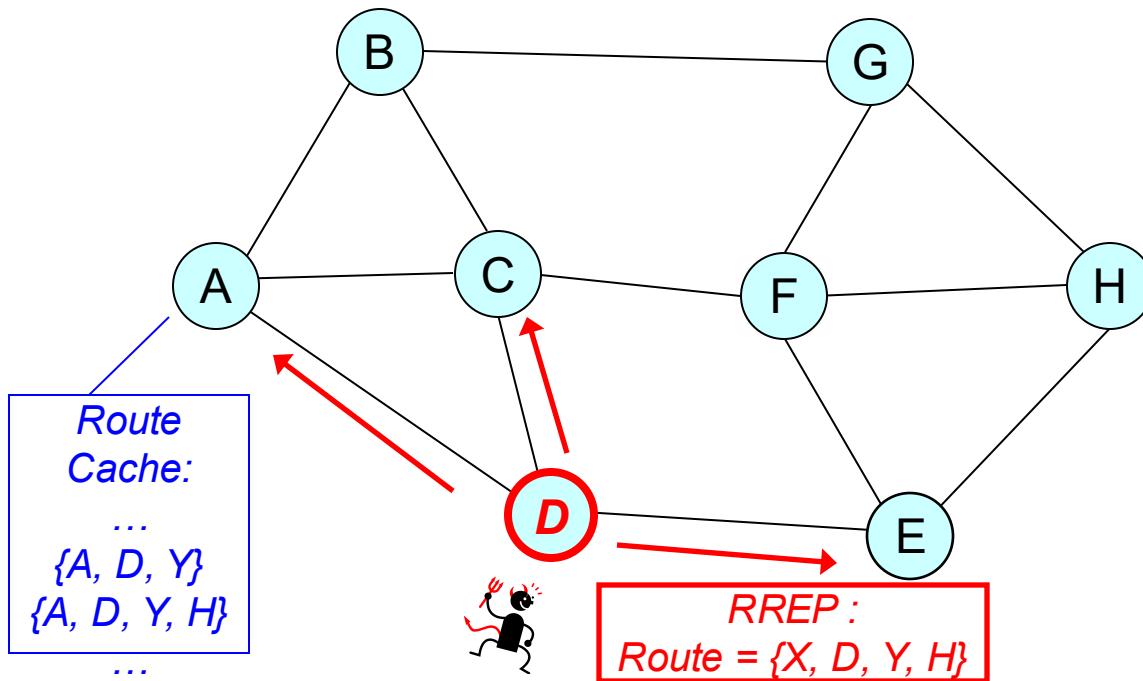
Attacking route discovery (cont'd)

- Modification of the route links, for example, in DSR



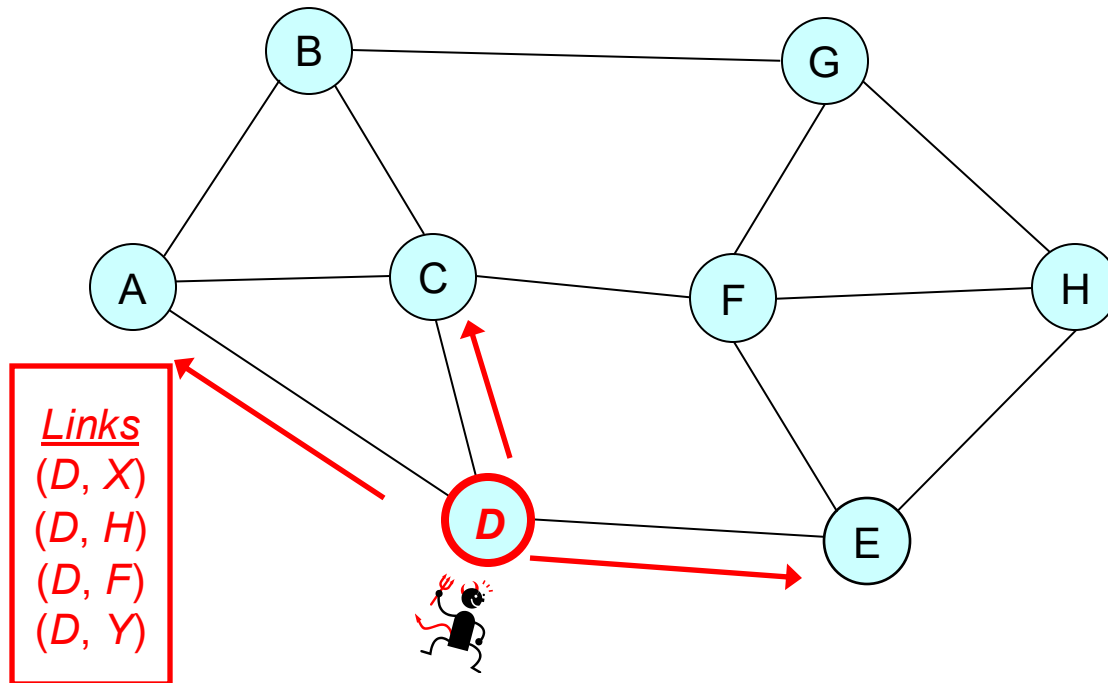
Attacking route discovery (cont'd)

- Abuse of the routing caching mechanism, for example, in DSR



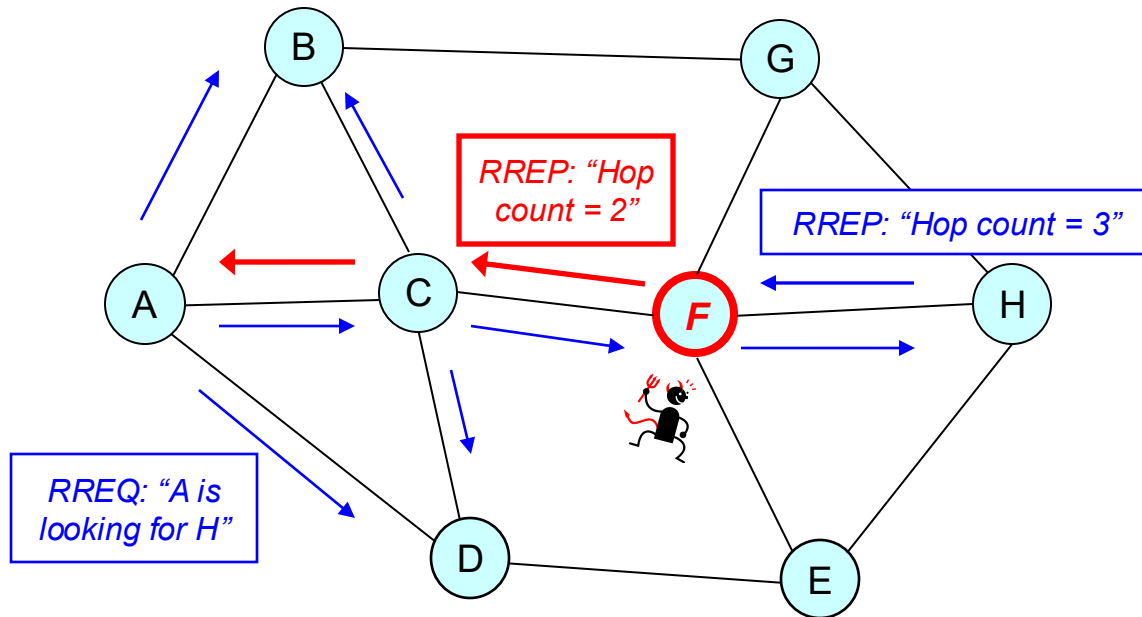
Attacking route discovery (cont'd)

- Disrupting a link state routing protocol, for example, in *OLSR*



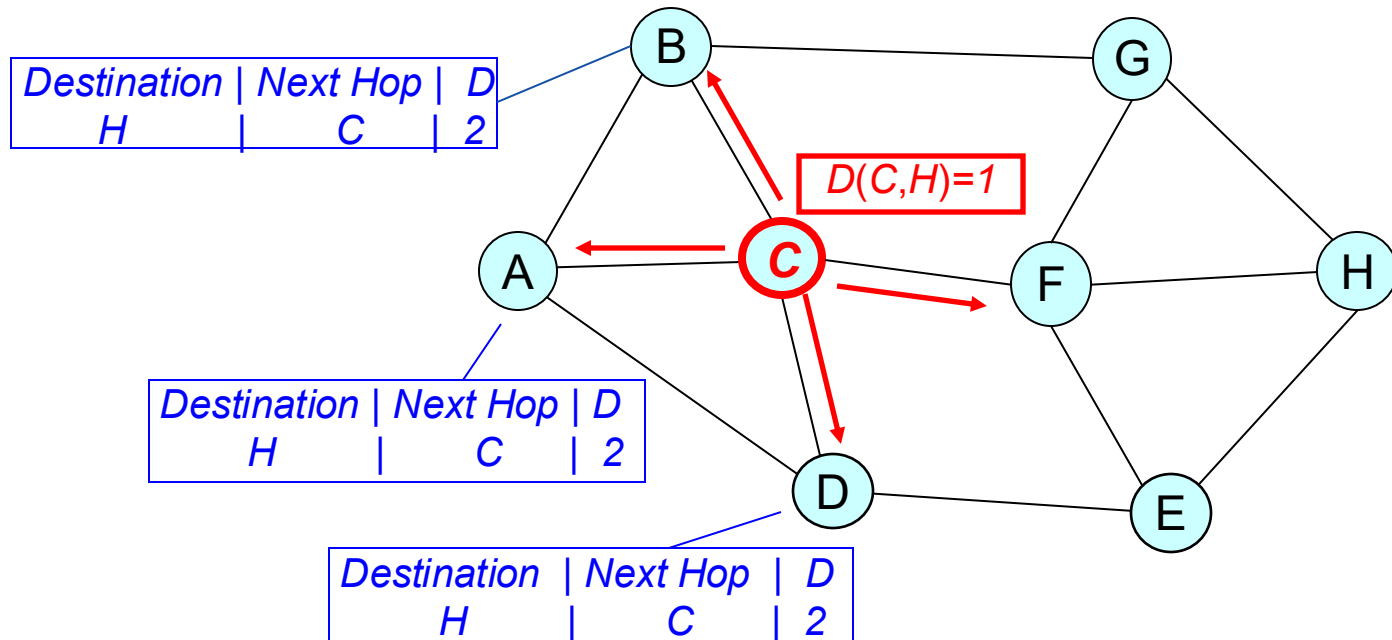
Attacking route discovery (cont'd)

- Disrupting distance vector routing, for example, in *AODV*



Attacking route discovery (cont'd)

- Disrupting distance vector routing, for example, in *DSDV*





Attacking route discovery (cont'd)

- Caution: none of the above-mentioned protocols (DSR, AODV, DSDV, OLSR) was designed with security in mind
- Many possible ways to attack the route discovery
- Outcome of attacks
 - Control communication
 - > *Become part of utilized routes*
 - > *Monopolize resources*
 - Disrupt communication
 - > *Degrade or deny*



Secure route discovery requirements

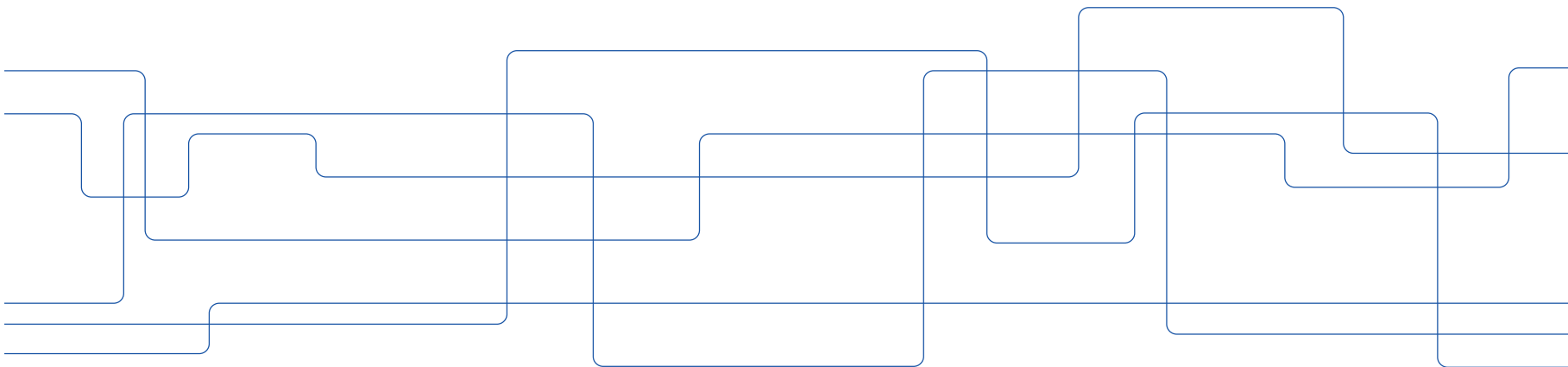
- What do we need a secure routing protocol to do?
- Network model
 - Capture the system characteristics
 - > *For example, dynamically changing topology*
- Specification
 - Define the properties of any candidate secure routing protocol independently of its functionality



Secure route discovery requirements (cont'd)

- We are interested in protocols that discover routes with the following two properties:
 - **(1) Loop-freedom:** an (S,T) -route is loop-free when it has no repetitions of nodes
 - **(2) Freshness:** an (S,T) -route is fresh with respect to a (t_1, t_2) interval if each of the route's constituent links is up at some point during the (t_1, t_2)
- Loop-freedom and freshness are relevant for both explicit and implicit route discovery, and both basic and augmented protocols

Secure Routing Protocol (SRP) Secure Link State Routing (SLSP) protocol

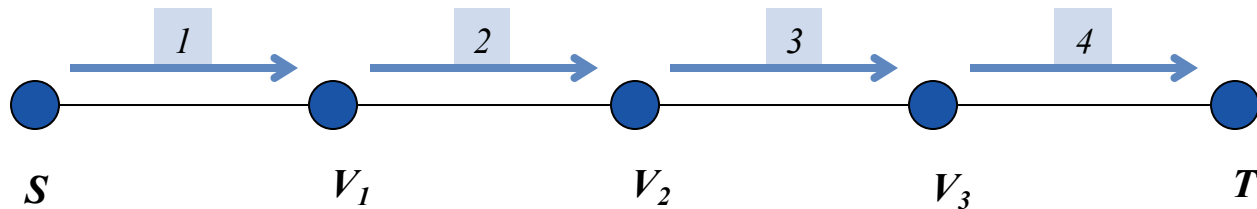




Secure Routing Protocol (SRP)

- Explicit basic route discovery
- Observation
 - It is hard to ‘know’ all nodes in the network, i.e., establish associations with all of them
 - Often infeasible and very costly
 - Especially in ‘open’ networks
- SRP assumptions
 - Secure neighbor discovery
 - Hop-by-hop authentication of all control traffic
 - End nodes (source, destination) ‘know’ each other
 - > *Can set up security associations*

SRP (cont'd)

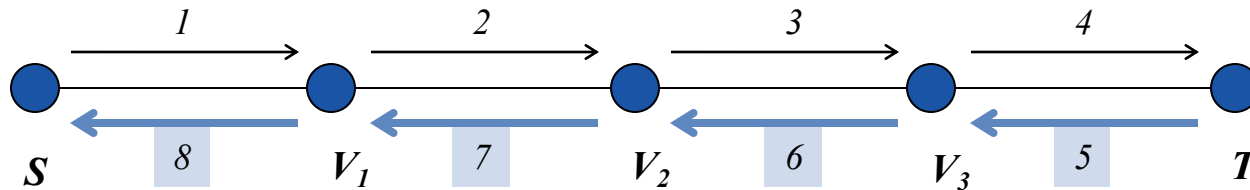


Route Request (RREQ):

$S, T, Q_{SEQ}, Q_{ID}, MAC(K_{S,T}, S, T, Q_{SEQ}, Q_{ID})$

1. S broadcasts *RREQ*;
2. V_1 broadcasts *RREQ*, $\{V_1\}$;
3. V_2 broadcasts *RREQ*, $\{V_1, V_2\}$;
4. V_3 broadcasts *RREQ*, $\{V_1, V_2, V_3\}$;

SRP (cont'd)



Route Reply (RREP):

$Q_{ID}, \{T, V_3, V_2, V_1, S\},$

$MAC(K_{S,T}, Q_{ID}, Q_{SEQ}, T, V_3, V_2, V_1, S)$

5. $T \rightarrow V_3 : RREP;$

6. $V_3 \rightarrow V_2 : RREP;$

7. $V_2 \rightarrow V_1 : RREP;$

8. $V_1 \rightarrow S : RREP;$



SRP (cont'd)

- Route requests verifiably reach destination
 - Intermediate node replies disabled
 - Aggressive caching of routing information disabled
- Route replies must trace back the paths traversed by route requests
- Intermediate nodes are not authenticated at the end nodes
- Dual route request identifier
 - Q_{ID} : random, used by the intermediate nodes
 - Q_{SEQ} : sequence number, used by the destination
 - The adversary cannot launch a “sequence number” attack



SRP (cont'd)

- Crucial to operate on top a secure neighbor discovery protocol
- Neighbor Lookup Protocol (NLP)
 - Secure neighbor discovery
 - Establish security associations between neighbors
 - Identify control traffic injected by each neighbor
 - Prevent attacks that misuse network addresses
 - > *IP spoofing*
 - > *Use of multiple identities*
 - > *MAC spoofing*
 - DoS protection
- Efficient mechanisms to discard spurious/ corrupted traffic at intermediate nodes
 - Replies relayed only if neighbors had previously forwarded the corresponding request



SRP (cont'd)

- Routes discovered by SRP in the presence of independent adversaries are fresh
 - t_1 is the point in time at which S transmitted a RREQ for T, and t_2 is the point at which S received the corresponding RREP
- In the presence of colluding adversaries SRP discovers 'weakly fresh' routes
 - A sequence of links, in general different than those in the discovered route were *up* at some point in (t_1, t_2)

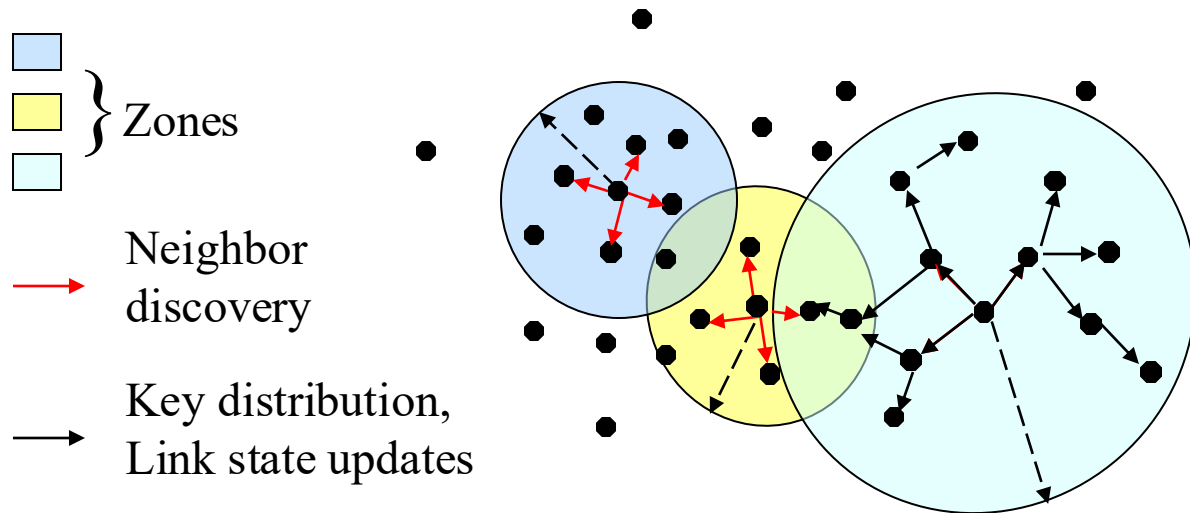


Secure Link State Protocol (SLSP)

- Secure Neighbor Discovery
 - Correct nodes discover only actual neighbors
- Periodic Link State Update (LSU) advertisements
 - Nodes distribute their discovered neighbors within an extended neighborhood, the *zone*
 - LSUs are signed
- Link state accepted *iff* reported by both incident nodes
- Nodes distribute their public key throughout the zone
- SLSP can adjust its scope with different zone radii

SLSP (cont'd)

- SLSP can adjust its scope, with different zone radii
- It can operate locally, combined with another global route discovery, or network-wide

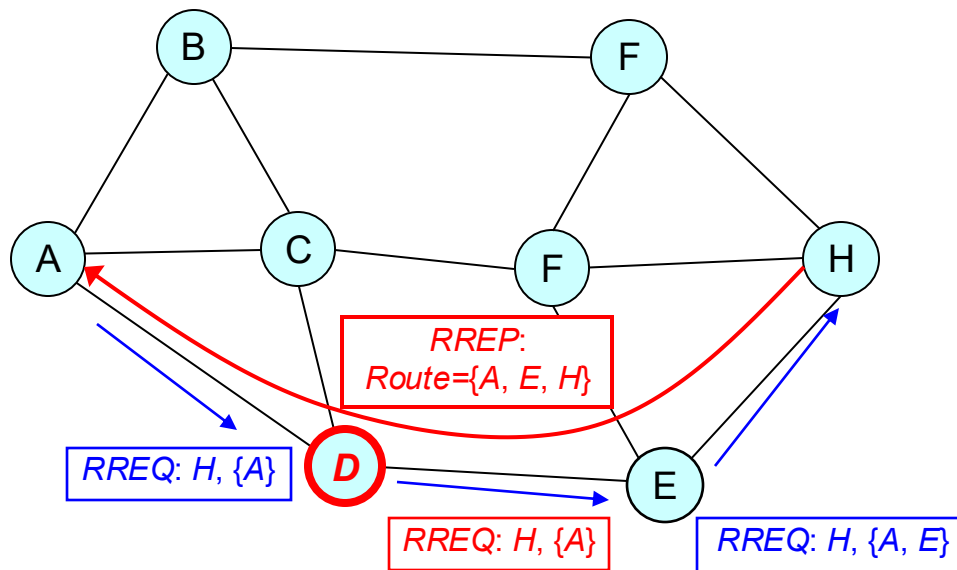




SLSP (cont'd)

- Keep the LSU propagation within the zone
 - Use a hash chain mechanism
 - $zone_radius = X_R = h^R(x_0)$
 - $hops_traversed = X_1 = h(x_0)$, $TTL = R-1$
 - After i hops ($i=R-TTL$), relay packet if:
 - > $i < R$, and
 - > $h^{R-i}(hops_traversed) == zone_radius$
 - $hops_traversed = H(hops_traversed)$
- Same idea can be applied in reactive routing, to perform an expanding ring search

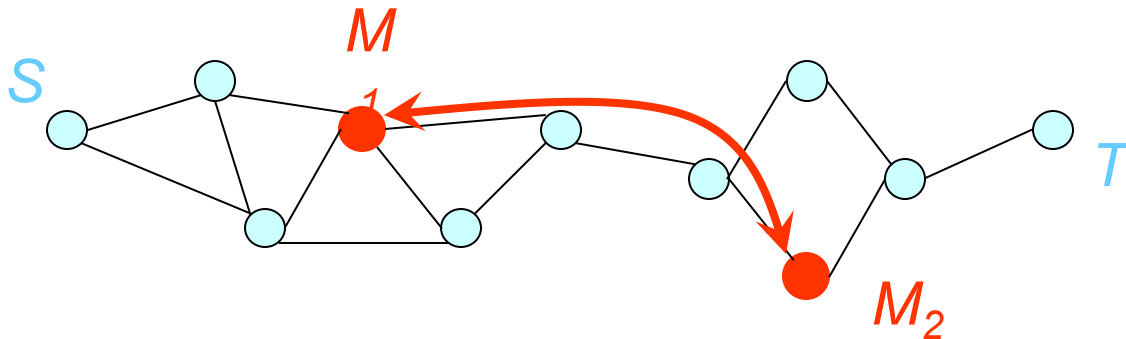
Attacking route discovery (cont'd)



- Adversary acting as a relay, 'creating' Byzantine links
- Secure neighbor discovery and hop-by-hop authentication can defeat this attack

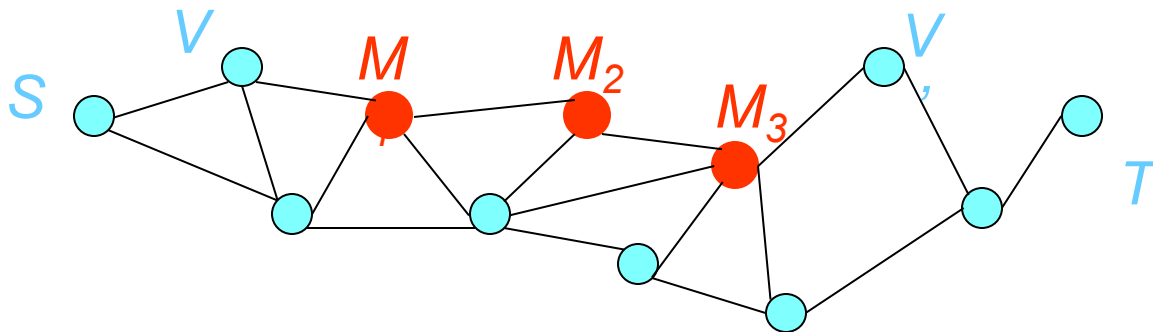
Attacking route discovery (cont'd)

- Tunneling Attack
 - Two colluding attackers: M_1 , M_2
 - M_1 encapsulates control traffic and forwards to M_2 and vice versa
 - Attackers seemingly follow the protocol with respect to their neighbors

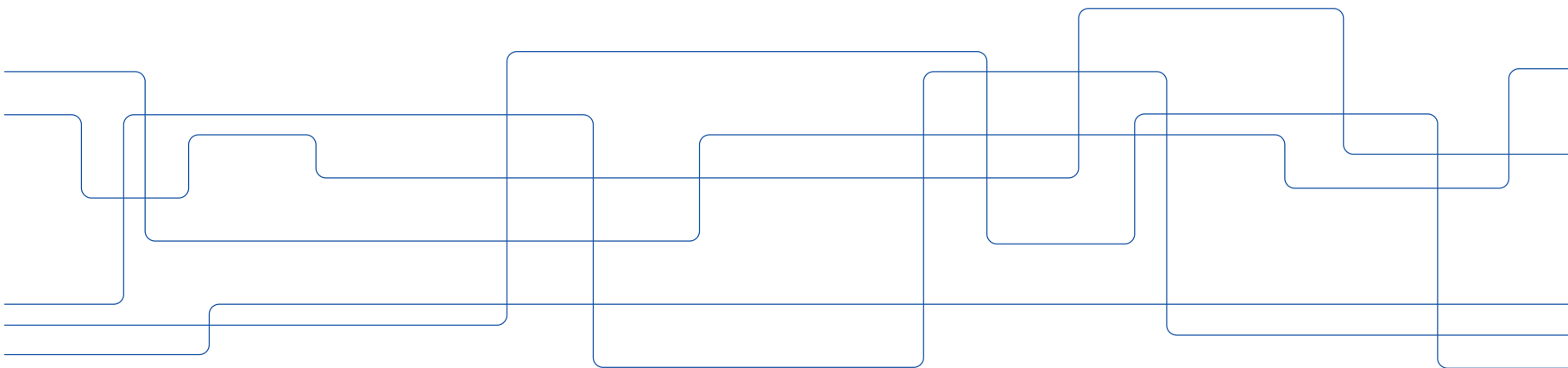


Attacking route discovery (cont'd)

- Multiple Colluding Attackers
 - M_1 and M_3 are seemingly correct to their neighbors, but they 'omit' protocol functionality when handling packets from M_2
 - Example: M_2 relays RREQ and RREP packets without appearing in the route discovery



Castor: Scalable Secure Routing for Ad hoc Networks



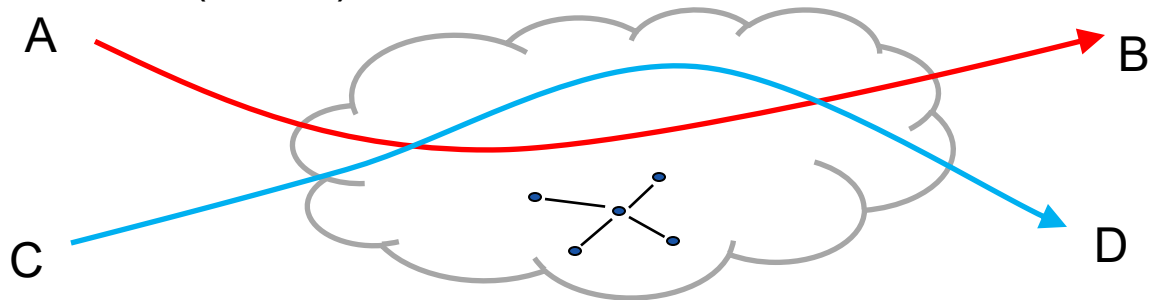


CASTOR

- **Continuously Adapting Secure Topology-Oblivious Routing**
 - Integrated route discovery and data communication
 - Localized, autonomous routing decisions
 - Simplest security associations (as for SRP+SSP/SMT)
 - Failure agnostic operation (as for SRP+SSP/SMT)
 - Bandwidth for security: “When in the dark, broadcast (but only for a short time)”
 - Outcome : Enhanced scalability and resilience

W. Galuba, **P. P.**, M. Poturalski, K. Aberer, Z. Despotovic, and W. Kellerer, “Castor: Scalable Secure Routing for Ad hoc Networks,” IEEE INFOCOM 2010

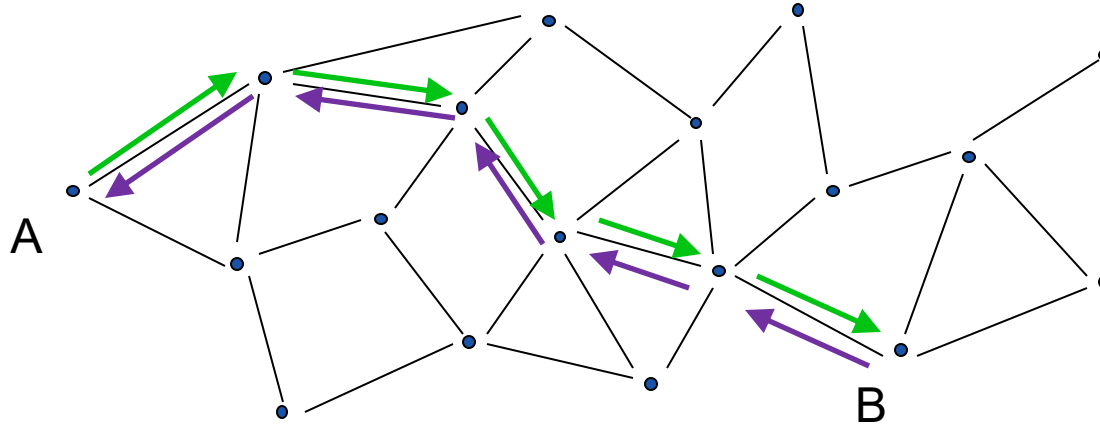
CASTOR (cont'd)



- End-to-end (EtE) and neighbor-to-neighbor (NtN) security associations
- EtE and NtN authentication only
- No routing packets, only data (PKTs) and acknowledgements (ACKs)
- In-network state updated based on PKTs and ACKs
- Crucial feature: flow isolation

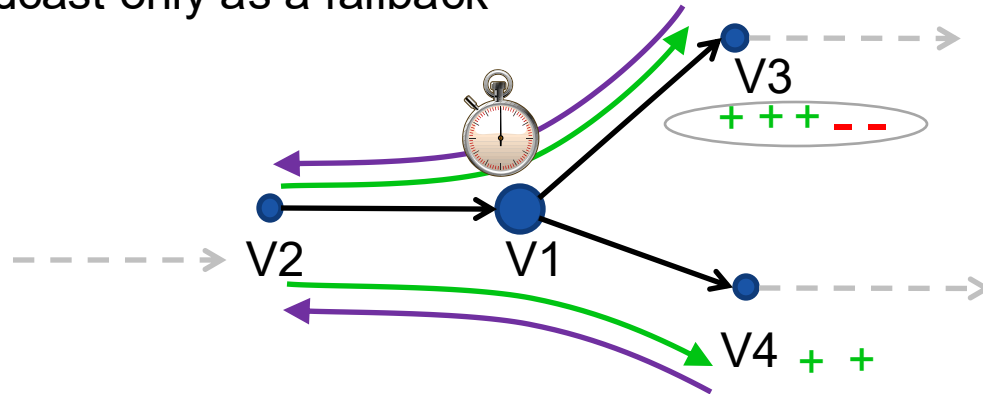
CASTOR (cont'd)

- Data in the **PKTs** payload
- **ACKs** follow the reverse path of **PKTs**



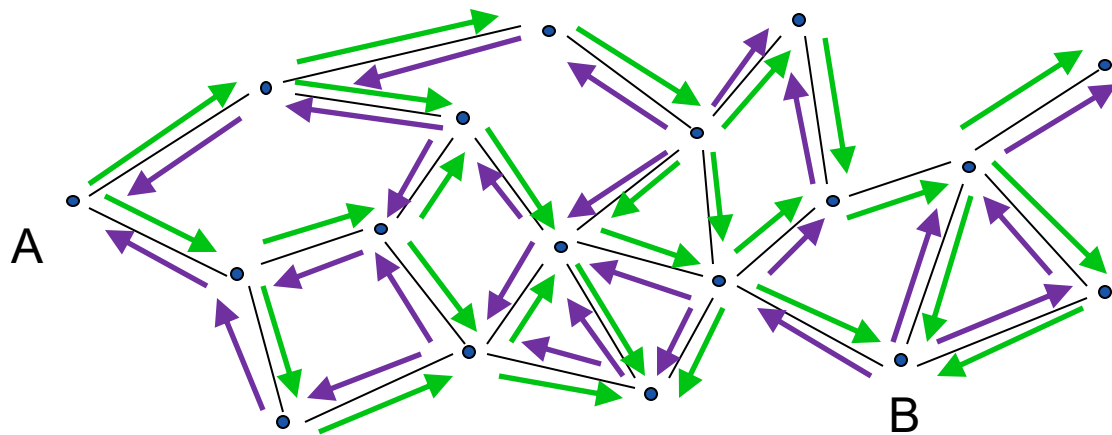
CASTOR (cont'd)

- Local learning from failures (of any type)
- Communication reliability estimators
 - Per flow and per neighbor
- Each node unicasts PKTs
- Broadcast only as a fallback



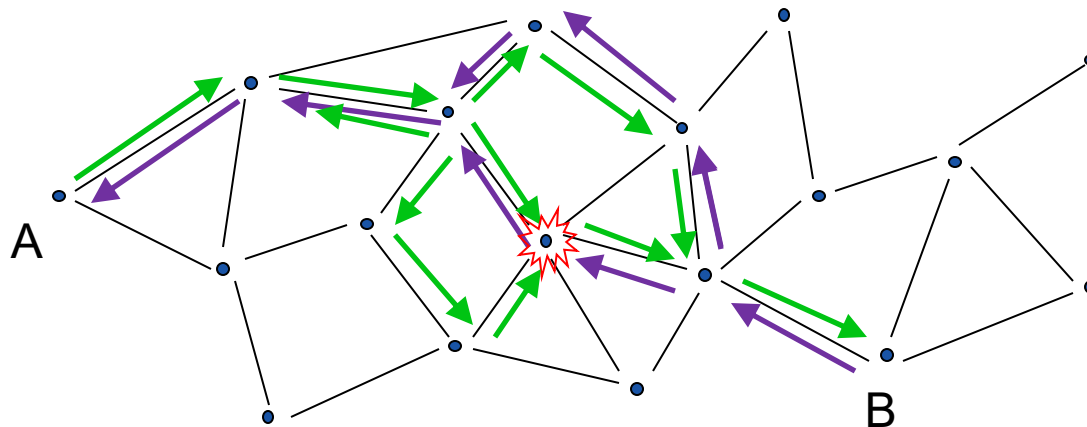
CASTOR (cont'd)

- Broadcast when there is no reliability history and no clear choice among neighbors
 - Each node decides autonomously for each PKT
 - If PKT was broadcast, the corresponding ACK is broadcast too



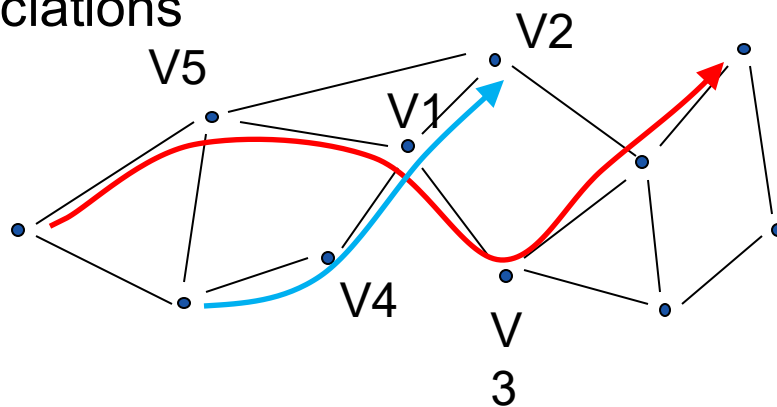
CASTOR (cont'd)

- Failure detection: No valid ACK
- Failure repair
 - Unicast the next PKT according to the local decision algorithm (in most cases and for most nodes)
 - Broadcast the next PKT (in few cases and for few nodes)



CASTOR (cont'd)

- Flow state isolation: “Must-do”
 - Example for V1 and the ‘blue’ and ‘red’ flows
- Otherwise adversaries could establish fictitious flows and block/trap legitimate flows
- Can be done with the assumed EtE (and NtN) security associations



Routing
State at V1

Node	Reliability	
V2	0.9	0.2
V3	0.6	0.95
V4	0.8	0.9
V5	0.1	0.8



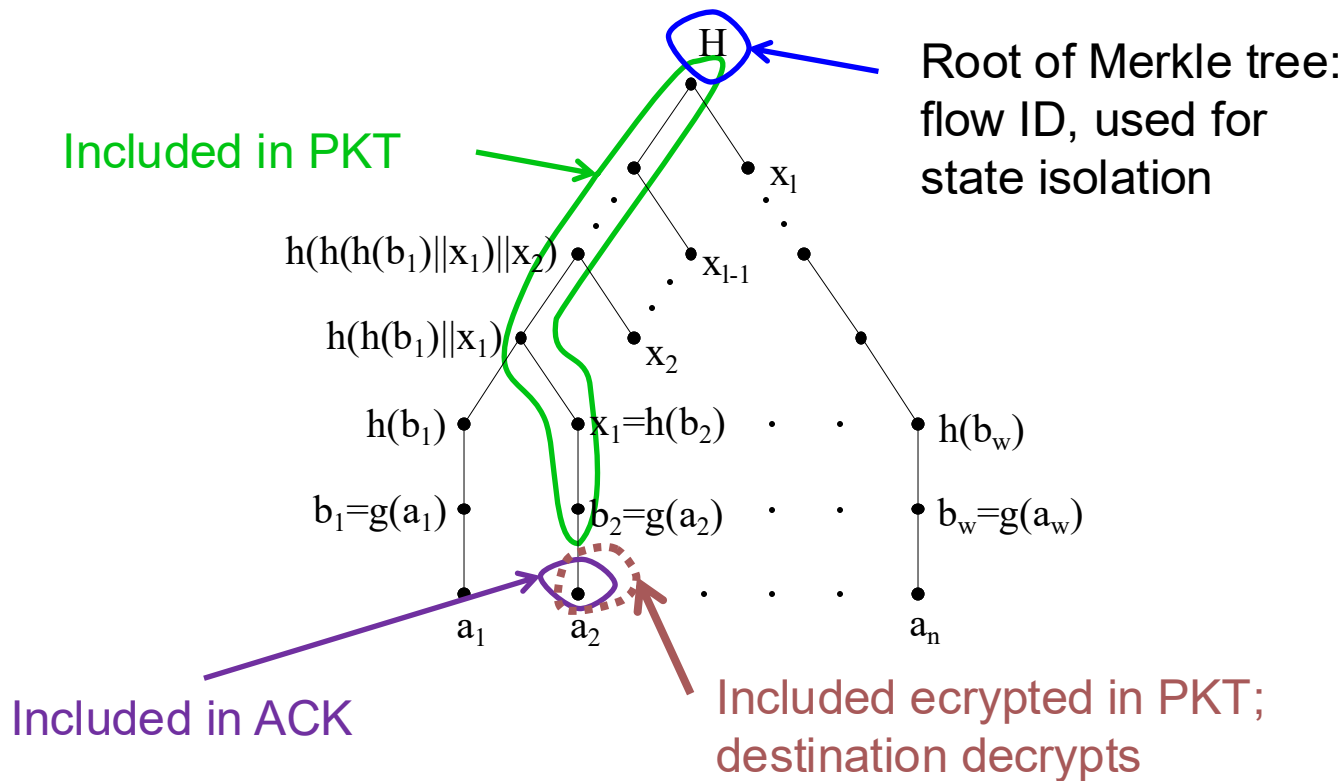
CASTOR (cont'd)

- Flow isolation
 - (Intermediate) nodes can recognize PKTs that belong to the same flow
 - Only the source of the flow can generate a valid next PKT
 - (Intermediate) nodes can match ACKs to PKTs
 - Only the destination can generate a valid ACK
 - No end-to-intermediate node authentication



CASTOR (cont'd)

Source node generates one Merkle tree per-flow

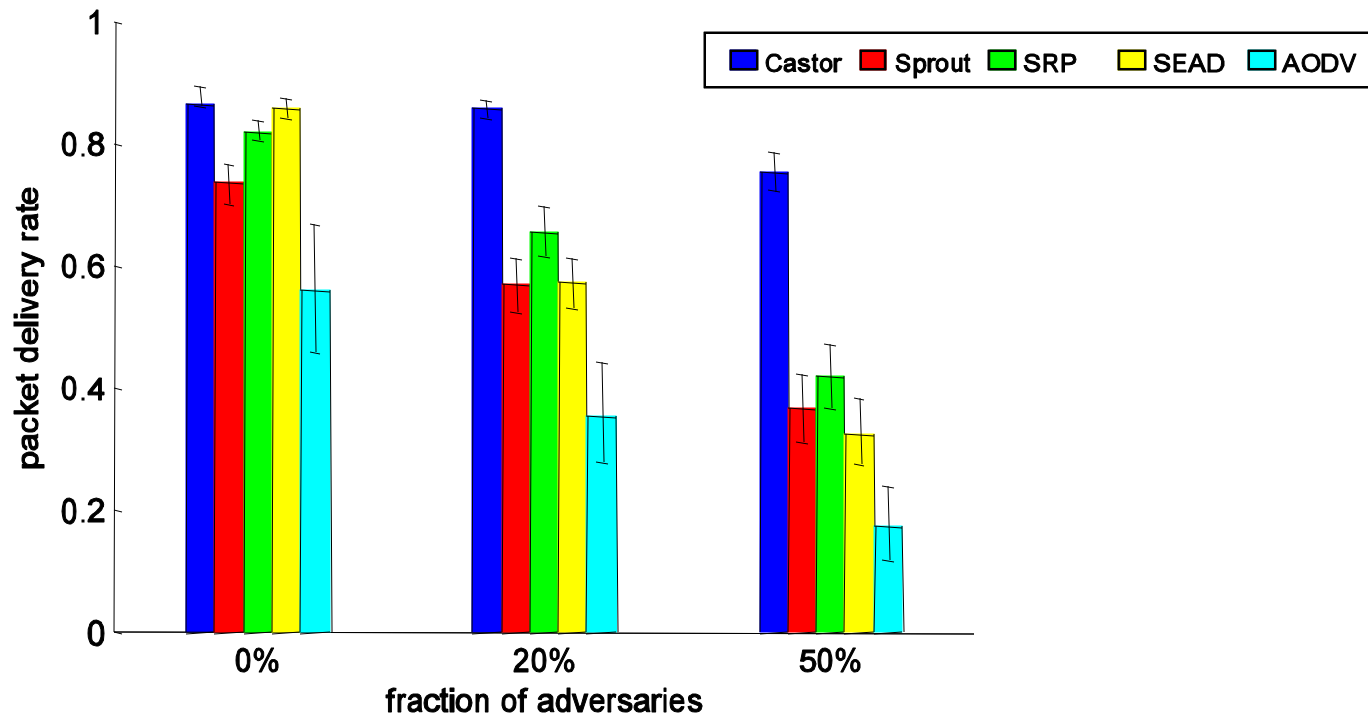


CASTOR (cont'd)

General			
plane size nodes		3km by 3km 100, placed uniformly at random, 10 radio neighbors on average	
MAC number of flows		802.11b at 1Mbps 5, source-destination disjoint	
flow rate		constant bit rate, 4 packets/s	
packet payload size		256 bytes	
Random waypoint mobility		Castor	
min. speed	1 m/s	γ	8
max. speed	20 m/s	δ	0.8
pause time	0s	T_{ACK}	500ms
SEAD			
periodic route update interval	5s		
SRP		Sprout	
α, β, δ	0.5	γ	1.25
r_s^{thr}	0	α_{pdr}	0.9
r_s^{max}	1	α_{rtt}	0.9

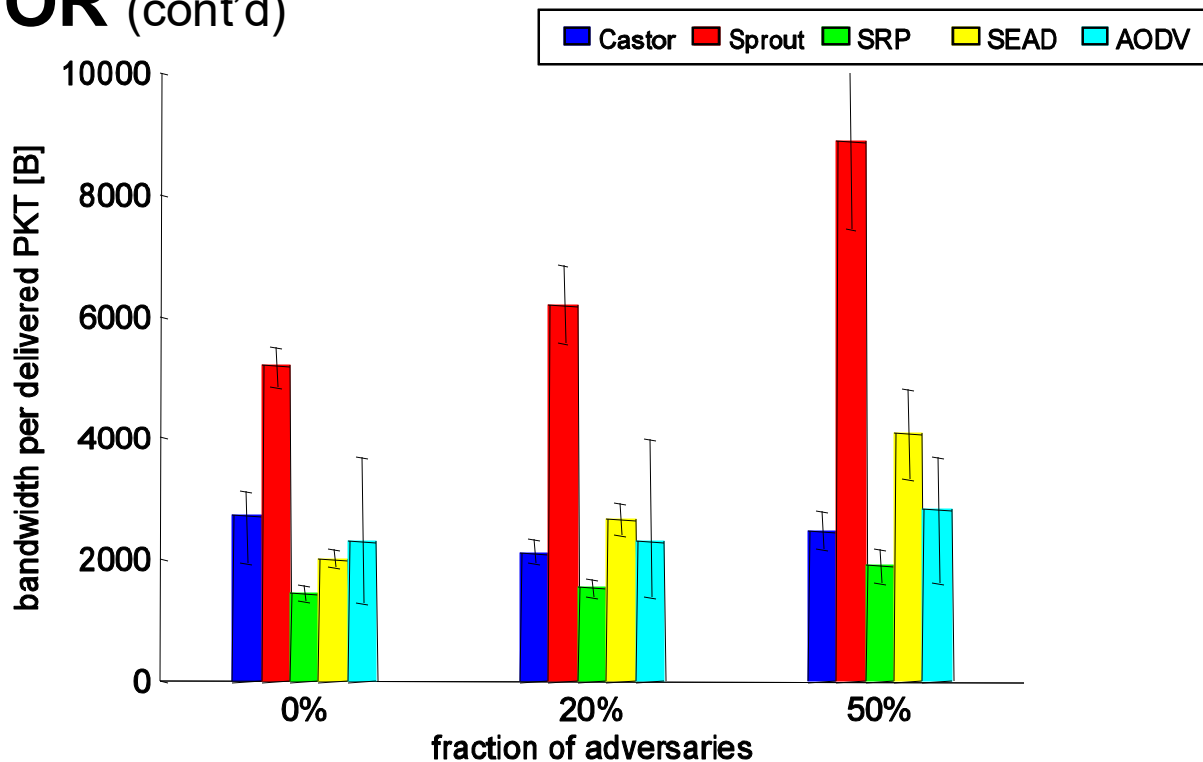
Note: “SRP” = “SRP + simplified SSP”

CASTOR (cont'd)



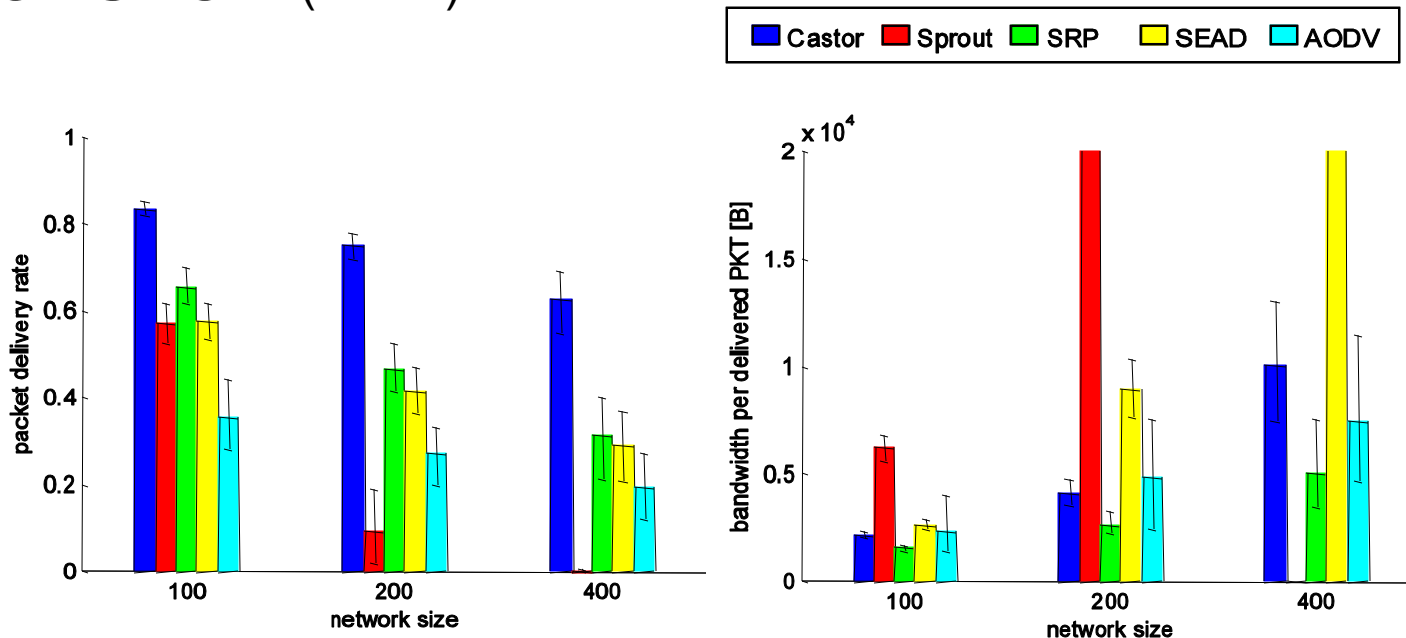
- Blackhole attack (dropping PKTs): PKT delivery ratio

CASTOR (cont'd)



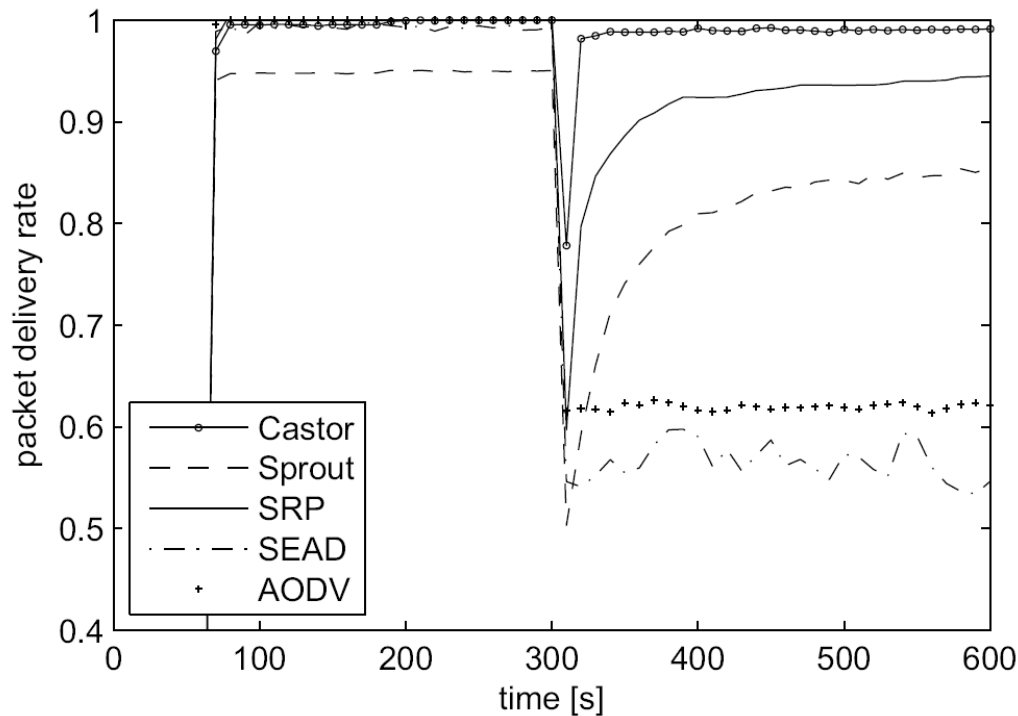
- Blackhole attack: Overhead (bandwidth utilization)

CASTOR (cont'd)



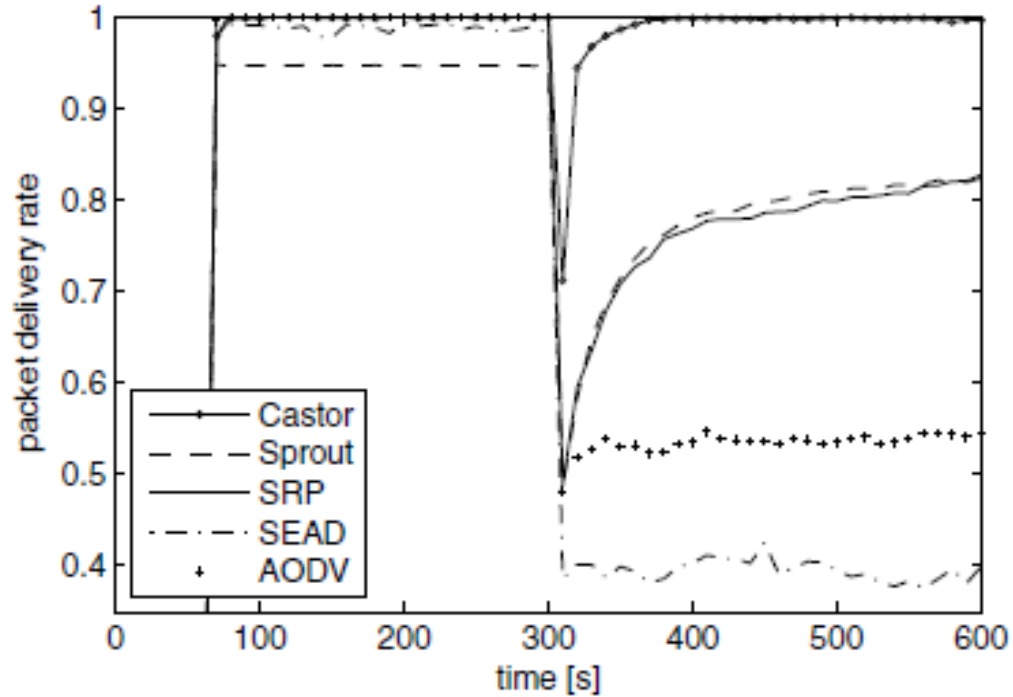
- Scalability: 100→200→400 node network
 - Mobility, 20% black hole nodes
 - PKT delivery and overhead

CASTOR (cont'd)

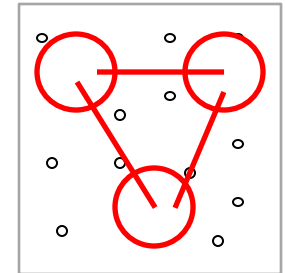


- Recovery from blackhole attacks

CASTOR (cont'd)



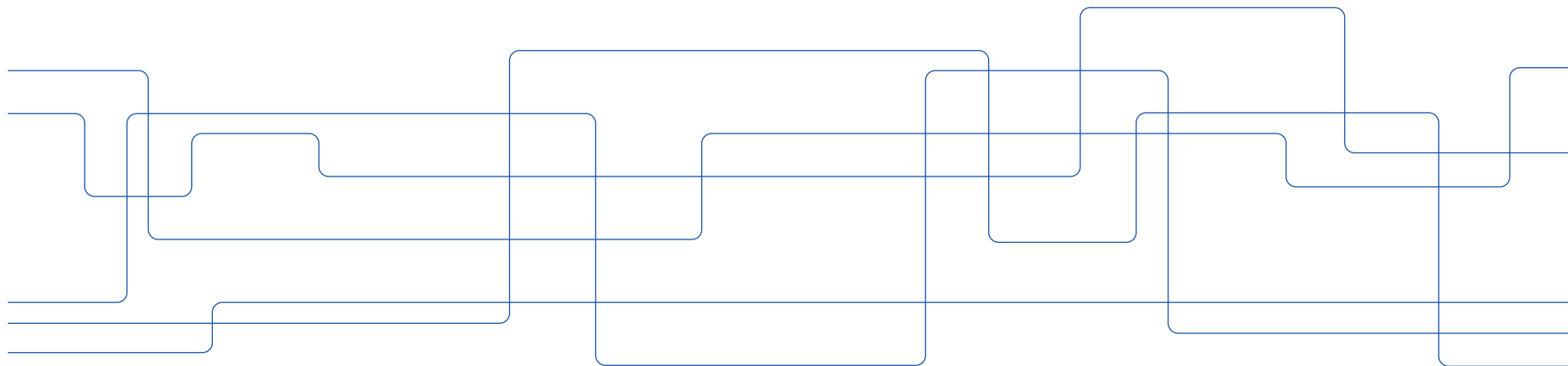
Recovery from wormhole attacks





Securing BGP

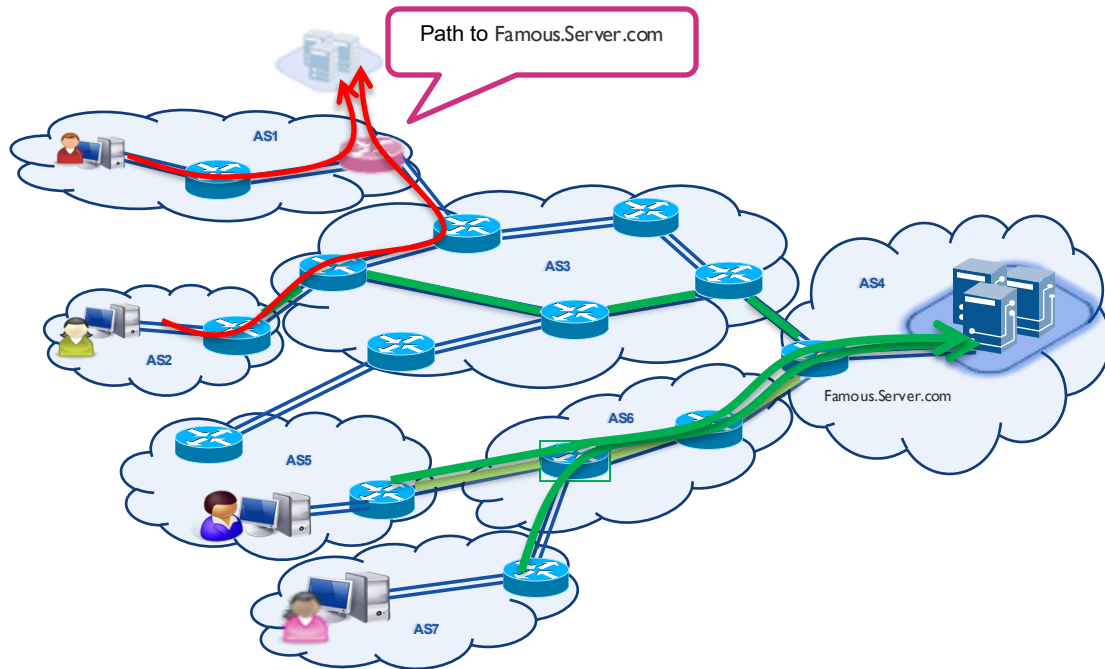
S-BGP, RPKI, BGPsec



Attacking Routing Protocols

BGP

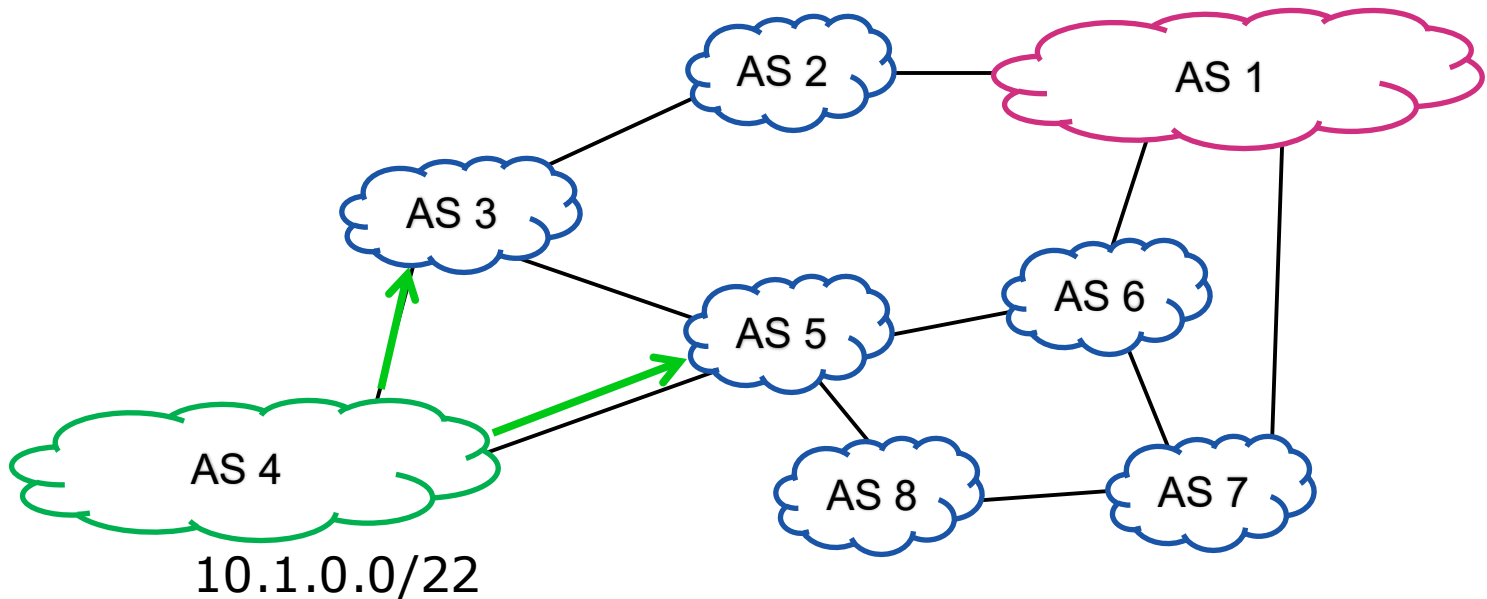
- Border Gateway Protocol (BGP): Inter-domain routing
- “The Internet’s **Biggest** Security Hole”
- Anyone can in principle originate advertisements to prefixes they have nothing to do with
- Discerning inappropriate prefix announcements: possible but hard
- Anyone with a BGP router could intercept data to any other network
- Stealthy version: data still forwarded to the destination



Attacking Routing Protocols (cont'd)

BGP

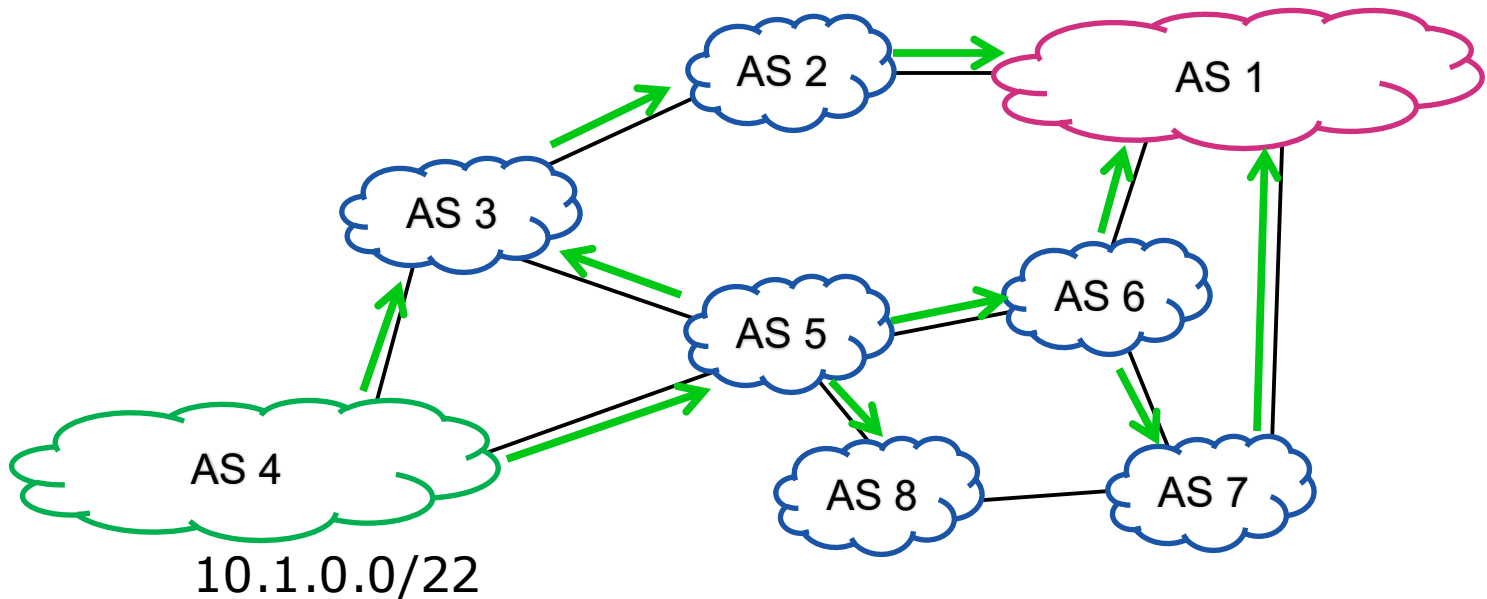
- AS 4 (the target) originates 10.1.0.0/22 and sends announcements to its neighbors



Attacking Routing Protocols (cont'd)

BGP

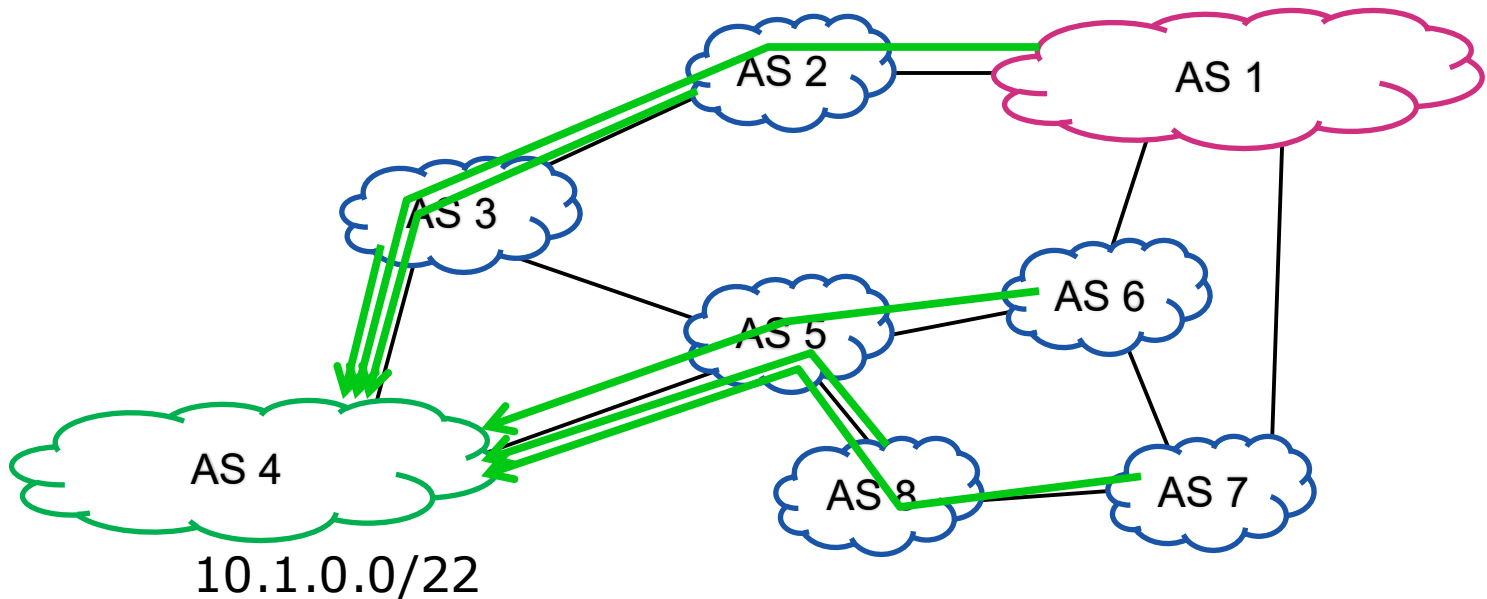
- The legitimate announcement propagates



Attacking Routing Protocols (cont'd)

BGP

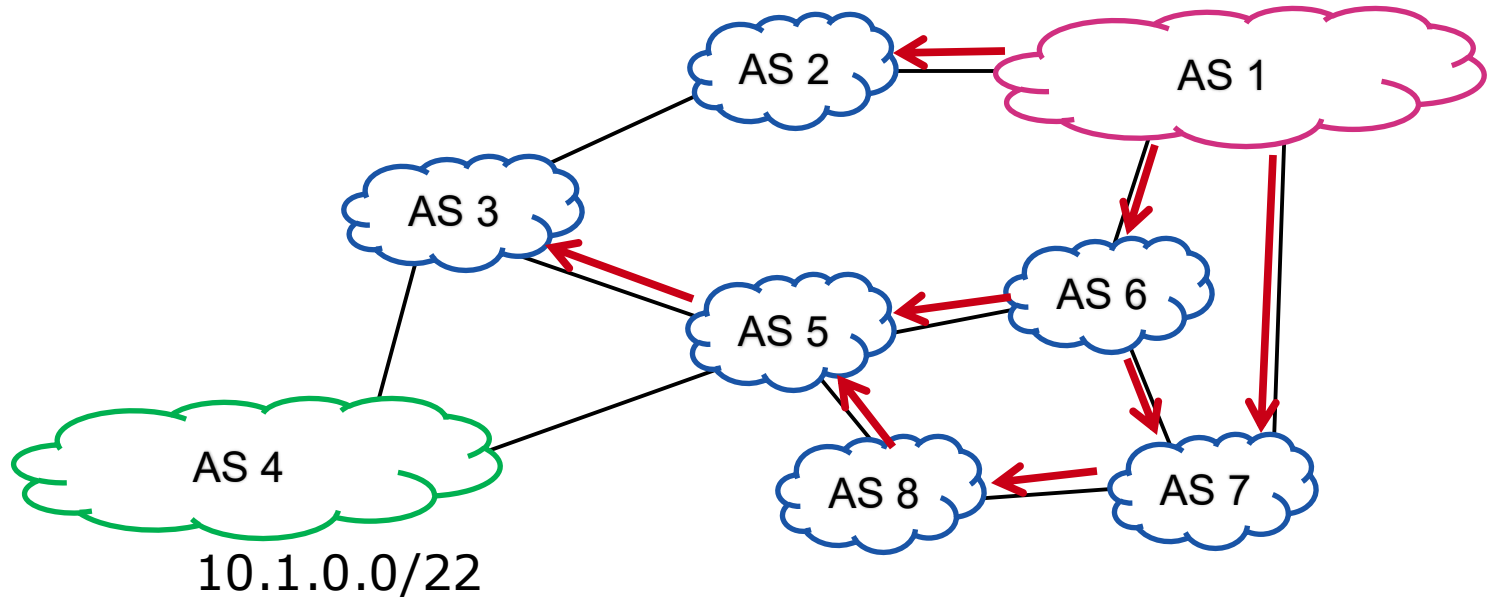
- View of Forwarding Information Base (FIB) for 10.1.0.0/22 after converging



Attacking Routing Protocols (cont'd)

BGP prefix hijack

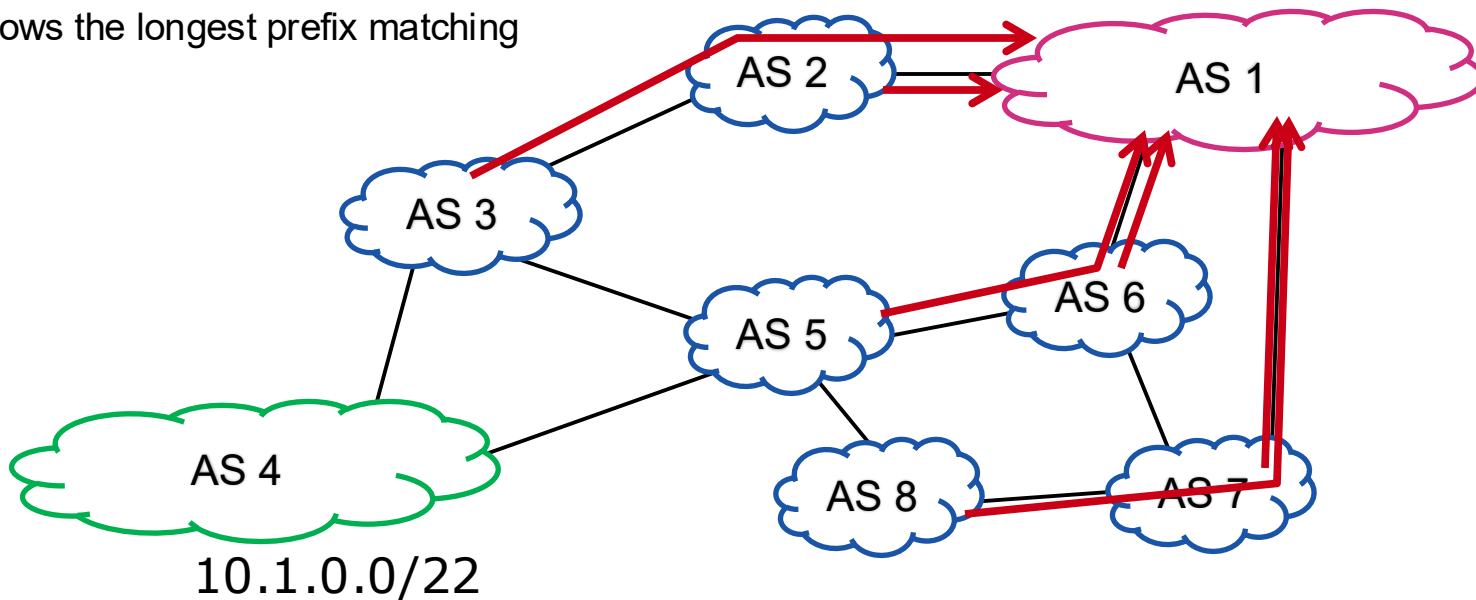
- AS 1 propagates a more specific route for AS 4, announcing 10.1.0.0/24



Attacking Routing Protocols (cont'd)

BGP prefix hijack

- Originating the same or a more specific prefix
 - Every AS picks the bogus route for that prefix
 - Traffic follows the longest prefix matching





Attacking Routing Protocols (cont'd)

BGP vulnerabilities

- Vulnerability of BGP
 - Origin authentication
 - > *Is the prefix owned by the AS announcing it?*
 - AS path authentication
 - > *Is AS path the sequence of ASes the BGP update traversed?*
 - AS path policy
 - > *Does the AS path adhere to the routing policies of each AS?*



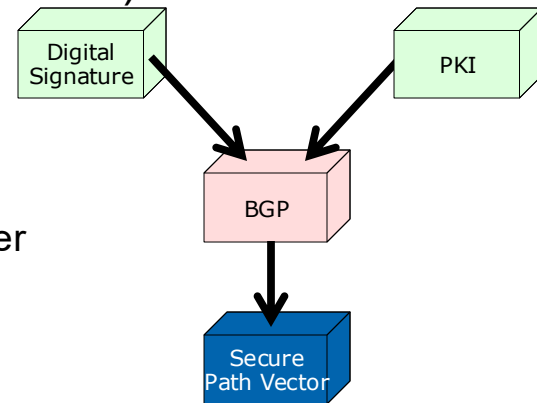
Securing BGP

- Secure BGP
- BGPsec
- RPKI

Securing Routing Protocols

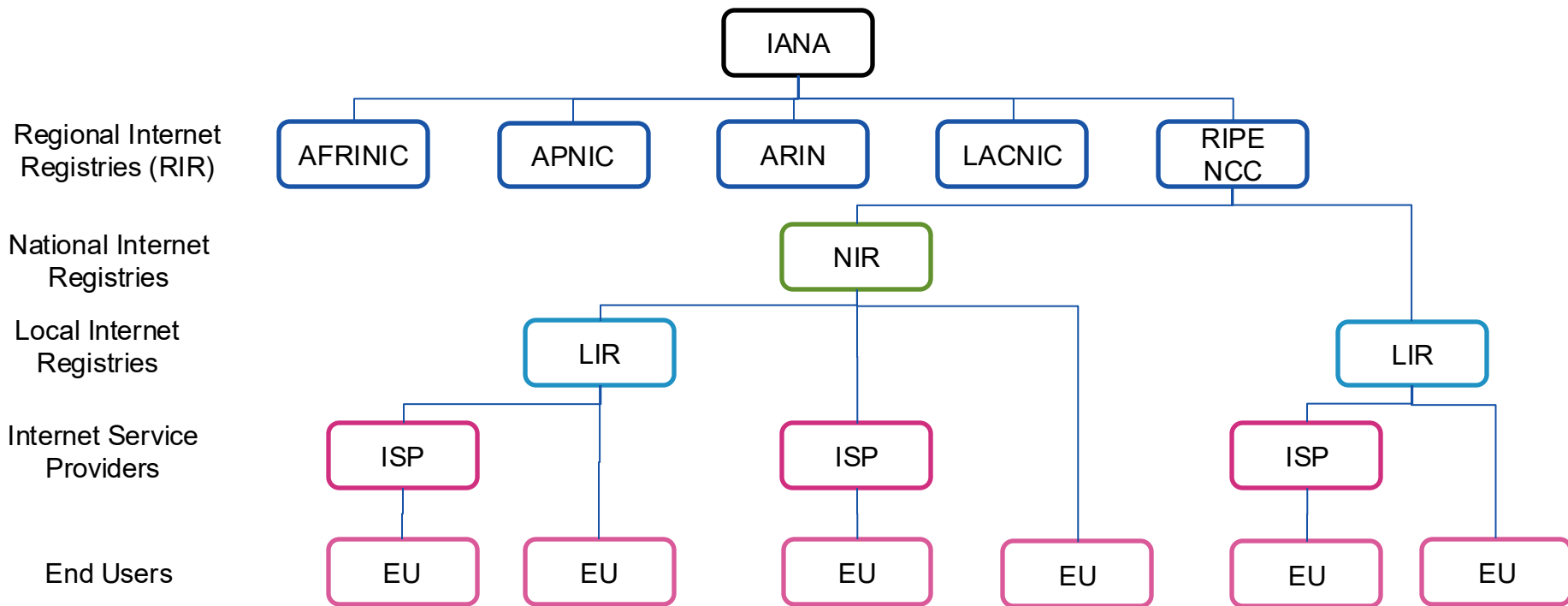
Secure BGP (S-BGP)

- S-BGP is the first comprehensive approach (BBN, 2000)
 - Centralized proposal based on a PKI
 - Signatures on every advertisement and element of the path
- Address-based PKI
 - Authenticate ownership for IP address block(s) and AS number
 - Use existing infrastructure (Internet registries etc.)
- Address attestation
 - Originating AS is authorized to advertise the destination block of addresses / prefix
- Route attestation
 - Next AS along the update path can use the AS Path, i.e., advertise the prefix



Securing Routing Protocols (cont'd)

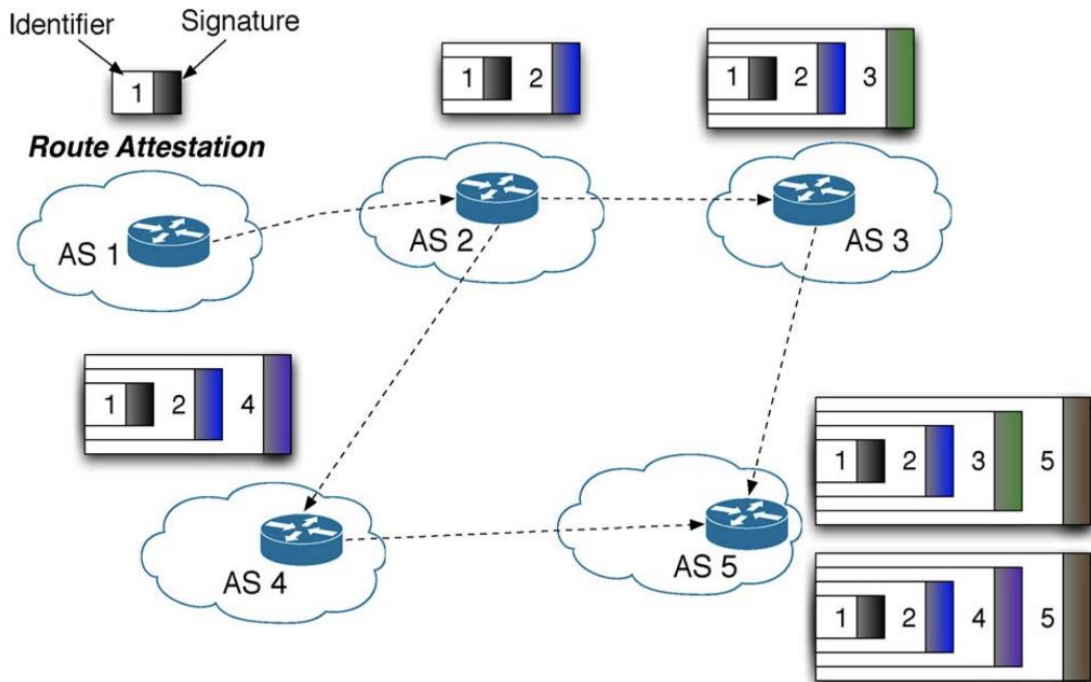
S-BGP Hierarchical Address Block PKI



Securing Routing Protocols (cont'd)

S-BGP attestations and updates

- Address attestation(s)
 - Maybe more than one; dep. on number of owners (sub-blocks) per advertised destination (block or Network Layer Reachability Information (NLRI))
- Route attestations
 - One per AS that relays the update
 - Identifies the next AS that can use Update
 - Also includes
 - > *The ID of the AS (its BGP speaker)*
 - > *The destination NLRI*
 - > *An expiration time*
 - > *Signature of the BGP speaker owner*

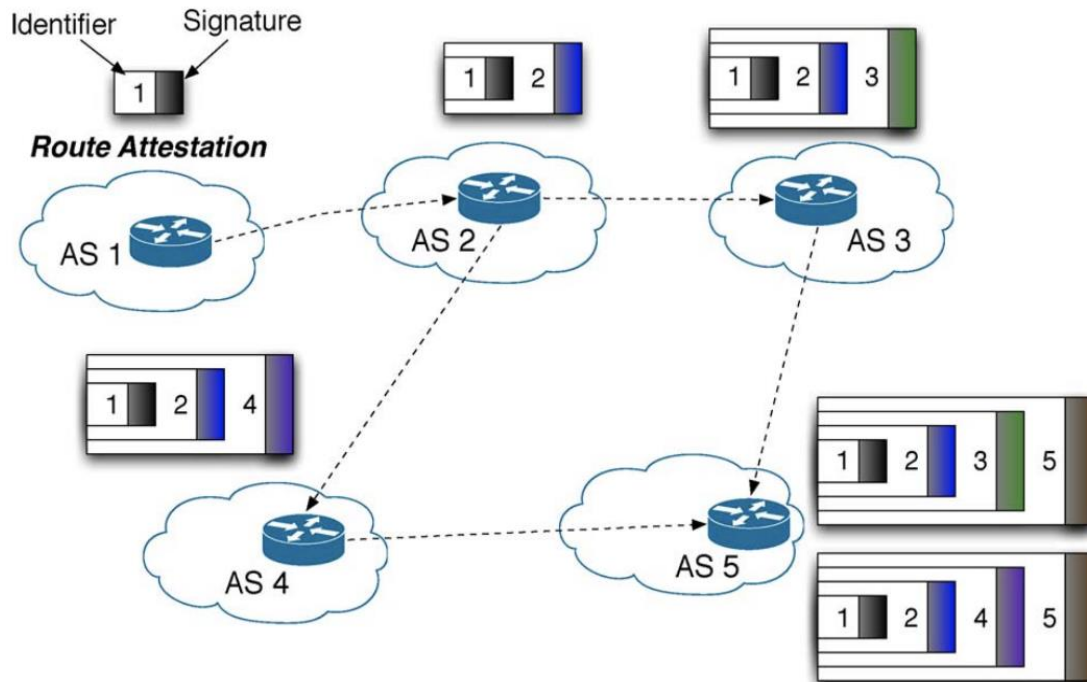


K. Butler, T. R. Farley, P. McDaniel and J. Rexford, "A Survey of BGP Security Issues and Solutions," in *Proceedings of the IEEE*, vol. 98, no. 1, pp. 100-122, Jan. 2010

Securing Routing Protocols (cont'd)

S-BGP attestations and updates – validation @ any receiving AS

- Address attestation
 - Address allocation certificate of the advertised address block owner
- Route attestation
 - Certificate of the signing AS
- Route update
 - Validated address attestation (one per owner)
 - One validated route attestation per AS in the received path
- All involved certificates must be valid (i.e., not revoked)



K. Butler, T. R. Farley, P. McDaniel and J. Rexford, "A Survey of BGP Security Issues and Solutions," in *Proceedings of the IEEE*, vol. 98, no. 1, pp. 100-122, Jan. 2010



Securing Routing Protocols (cont'd)

S-BGP – Challenges

- Public Key Infrastructure
- Security overhead, transmission and computation
- Handling changes
 - Route withdrawals, change in ownership
- Message suppression: BGP router 'choosing' not to relay control traffic
- Connectivity information disclosure
- Replay attacks
 - Premature re-advertisement of withdrawn routes within the lifetime a route attestation
- Incorrect following/application of policies and rules
 - Routers are not strictly bound by the protocol



Securing Routing Protocols (cont'd)

Deploying security for BGP

- S-BGP was not deployed
 - Implemented in a test environment
 - Design ideas proposed were highly influential
- Even without cryptography and PKI: prefix advertisements need to be validated
 - Internet Routing Registry (IRR): <http://www.irr.net/>
 - > *Addition of prefix information*
 - > *Filtering announcements*
 - Not globally adopted, hard to have accurate information
- Internet Engineering Task Force (IETF) Secure Inter-Domain Routing (SIDR) Working Group (WG)
 - Prefix validation
 - > *Resource Public Key Infrastructure (RPKI)*
 - > *Route Origin Authorization (ROA)*
 - AS path
 - > *BGPsec*

Securing Routing Protocols (cont'd)

Resource Public Key Infrastructure (RPKI)

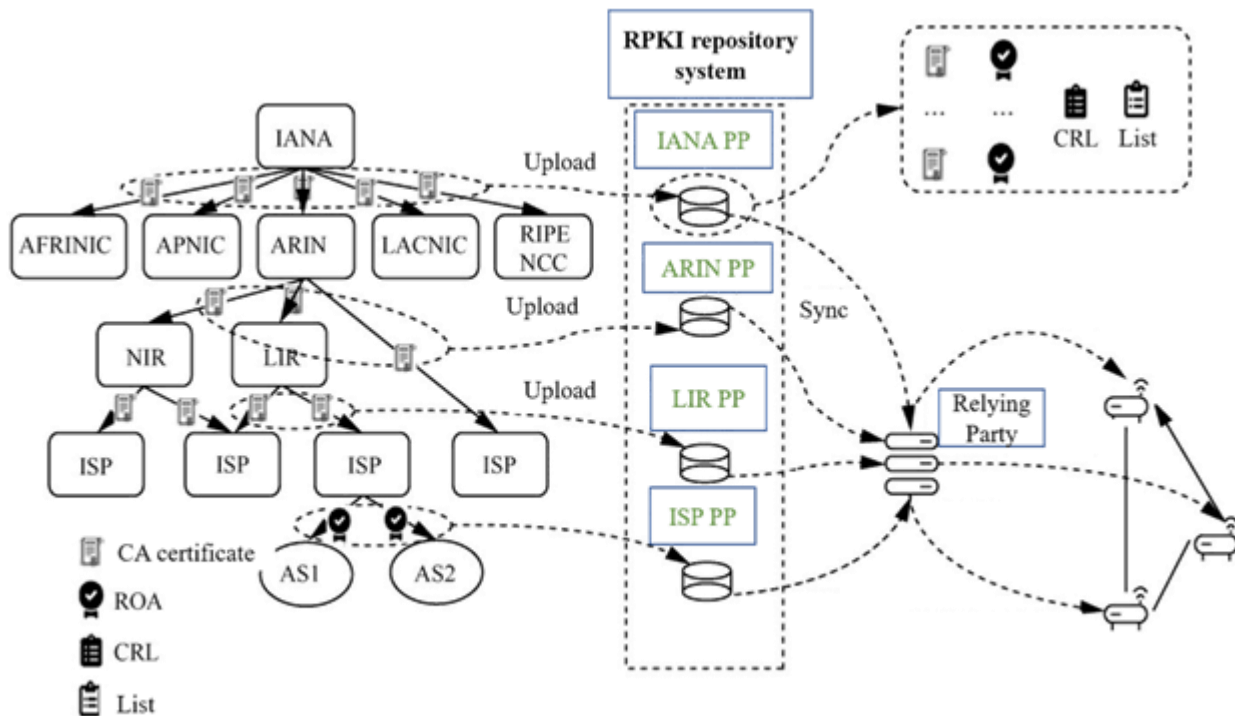
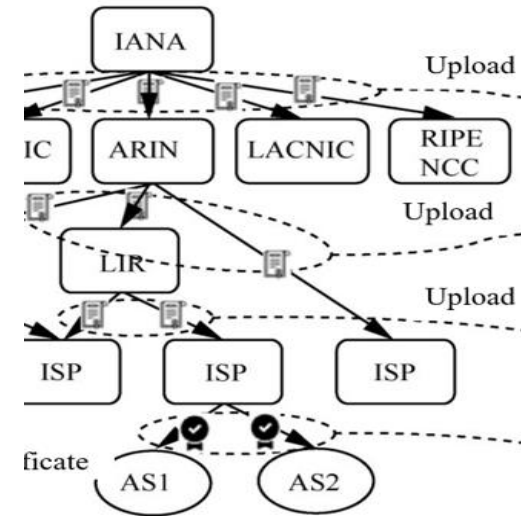


Figure by Di Wu, adapted from: Yingying SU, Dan LI, Honglin YE. Resource public key infrastructure RPKI: status and problems

Securing Routing Protocols (cont'd)

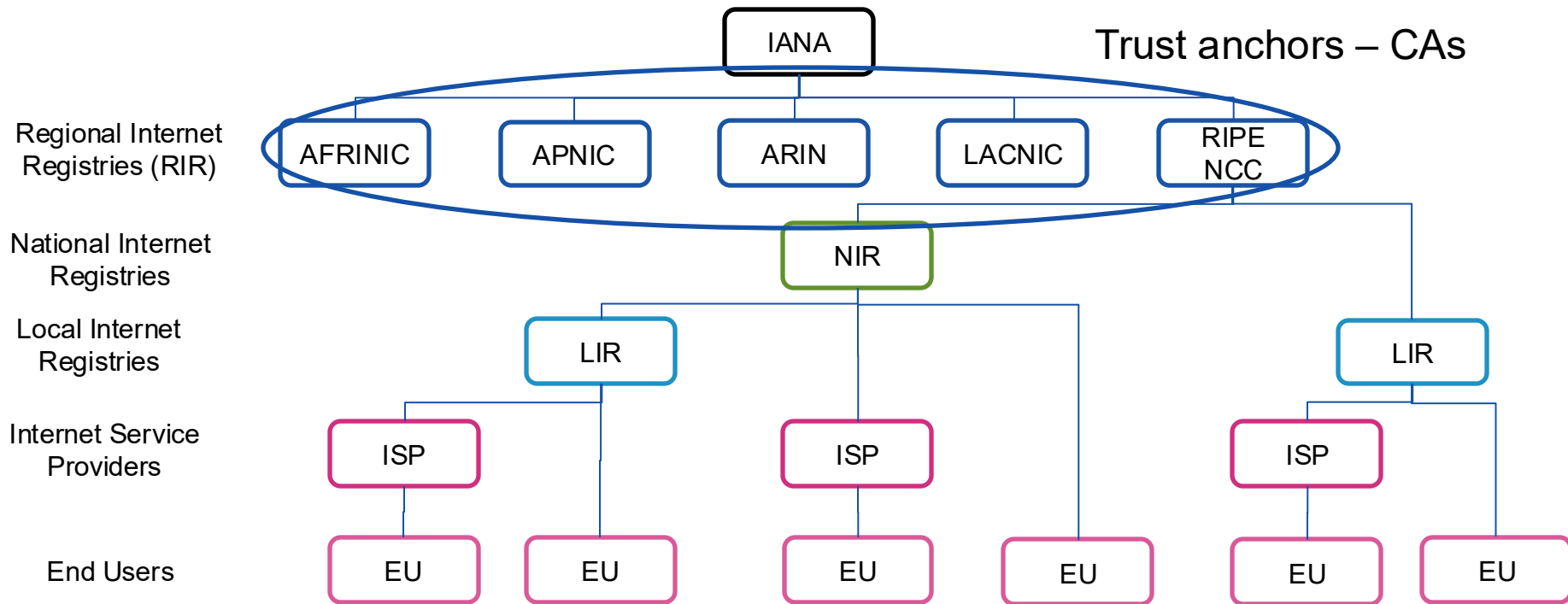
RPKI

- Internet Assigned Numbers Authority (IANA) root of trust
- Public key infrastructure (PKI)
- Objective: prevent prefix hijacking attacks
- Approach
 - Announce resources (IP prefixes) and which entity has the right to announce them
 - Make this information broadly available in the form of certificates
 - All networks can access and validate it in order to use it
 - BGP does not need to change but can filter out unauthorized announcements



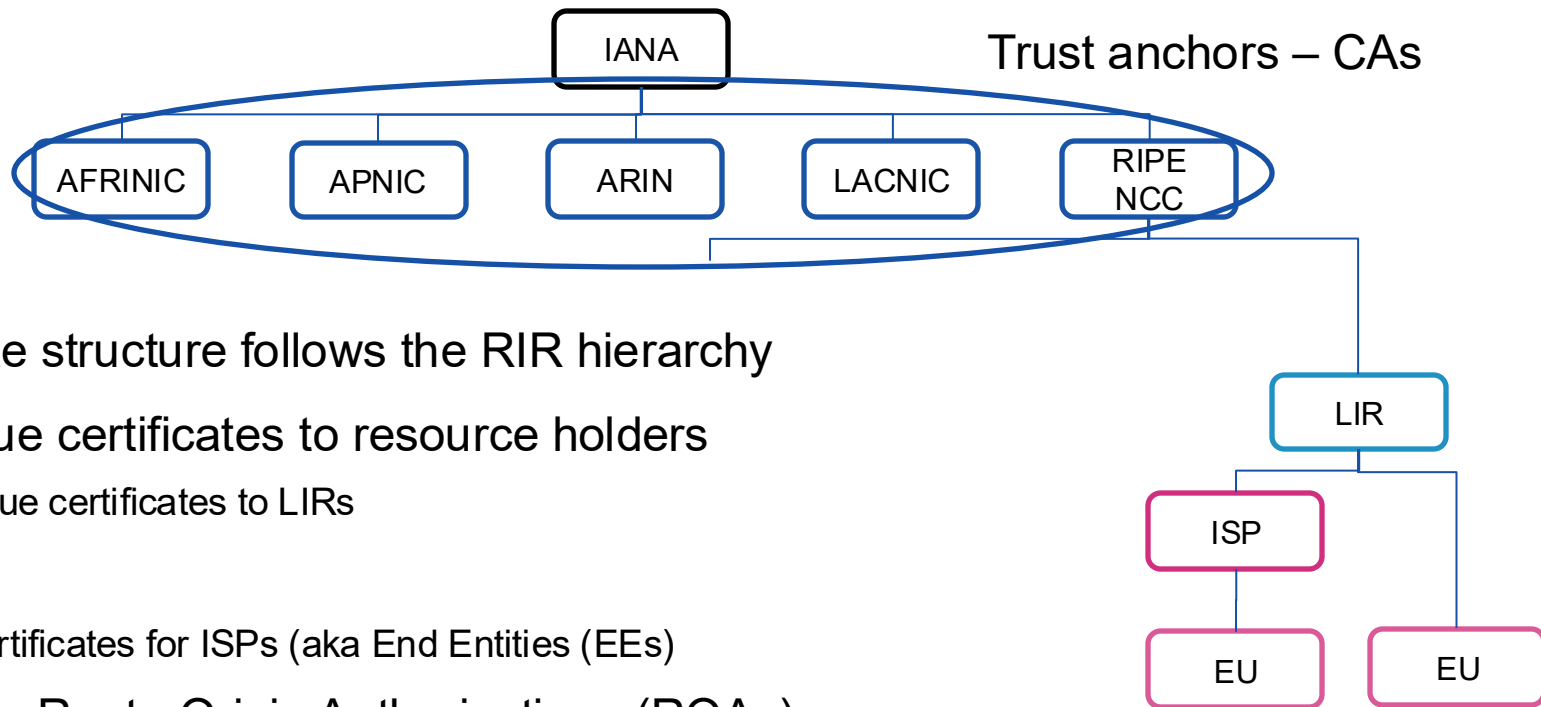
Securing Routing Protocols (cont'd)

RPKI



Securing Routing Protocols (cont'd)

RPKI



- Certificate structure follows the RIR hierarchy
- RIRs issue certificates to resource holders
 - RIRs issue certificates to LIRs
- LIRs
 - Issue certificates for ISPs (aka End Entities (EEs))
- EEs issue Route Origin Authorisations (ROAs)

Securing Routing Protocols (cont'd)

RPKI

For all recourses

Root Certificate

Public Key
Digital Signature

Root certificate

- Self-signed
- RIRs use root certificate to sign CAs' certificates

For all recourses

CA Certificate

Public Key
Digital Signature

CA certificate

- Binding the Internet Number Resources (INRs), including ASNs and IP prefixes to a public key

End Entity (EE) certificate

- Signing non certificate objects, notably ROAs

EE Certificate

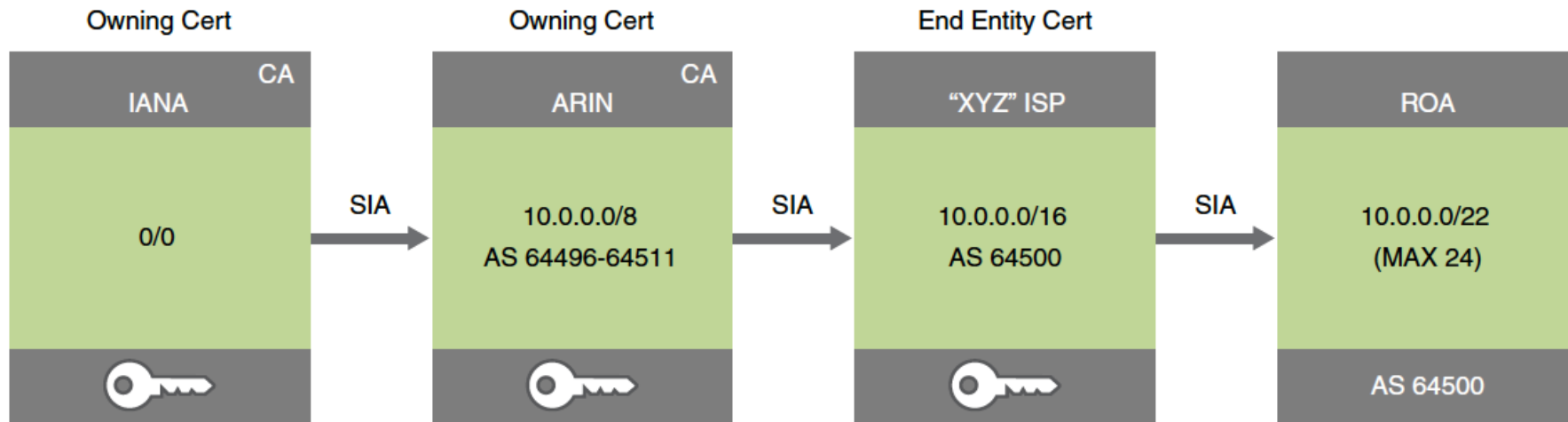
Public Key
Digital Signature

ROA

Digital Signature

Securing Routing Protocols (cont'd)

RPKI



https://www.juniper.net/documentation/en_US/release-independent/nce/information-products/pathway-pages/nce/nce-187-bgp-rpki-tn.pdf

- Route Origin Authorization (ROA)
 - Authorizing an AS to announce certain IP prefixes, signed by EE certificate

Securing Routing Protocols (cont'd)

RPKI

- Route Origin Authorization (ROA)
 - Authorizing an AS to announce certain IP prefixes, signed by EE certificate
- Max Length
 - The max prefix length that can be advertised
- Origin AS
 - The ASN expected to originate the BGP announcement
- Alternative ways to issue ROAs

ROA

Prefix 137.29.138.0/24

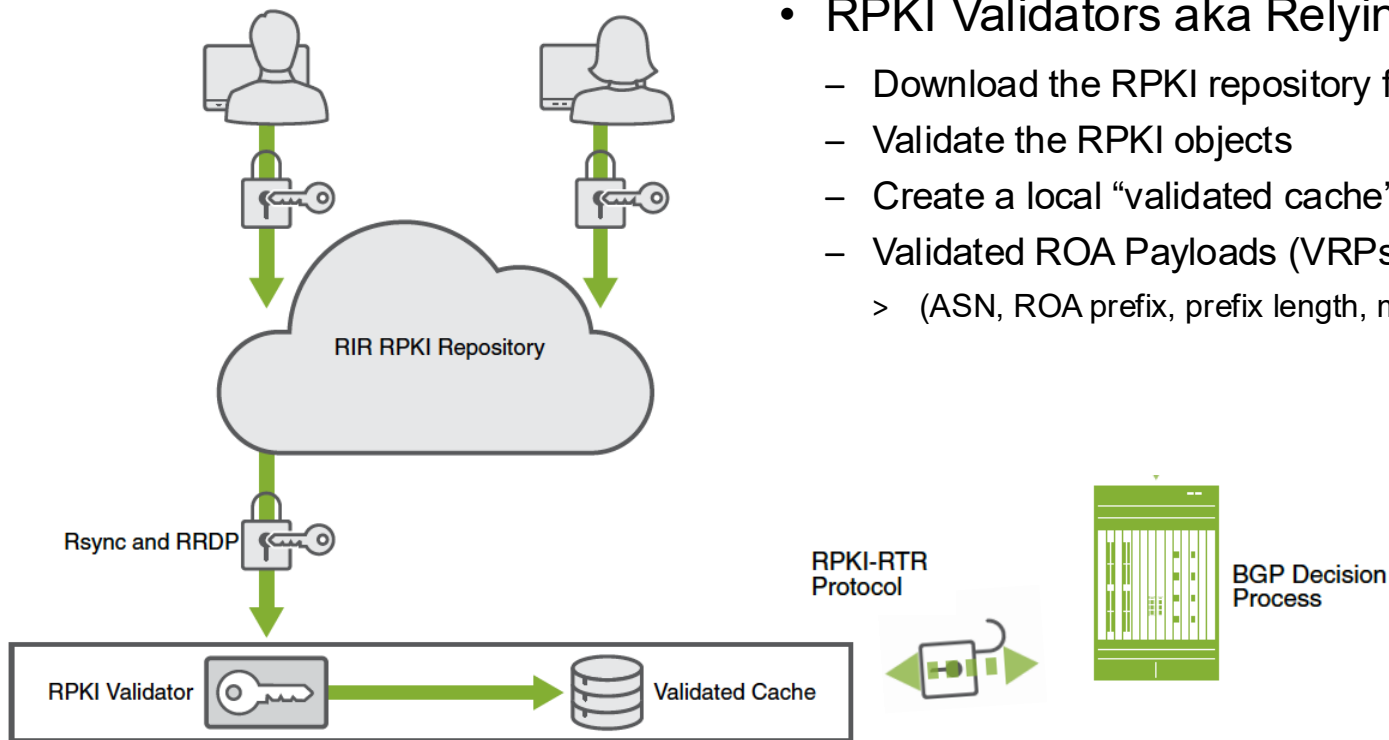
Max Length /24

Origin AS AS6589



Securing Routing Protocols (cont'd)

Route origin validation

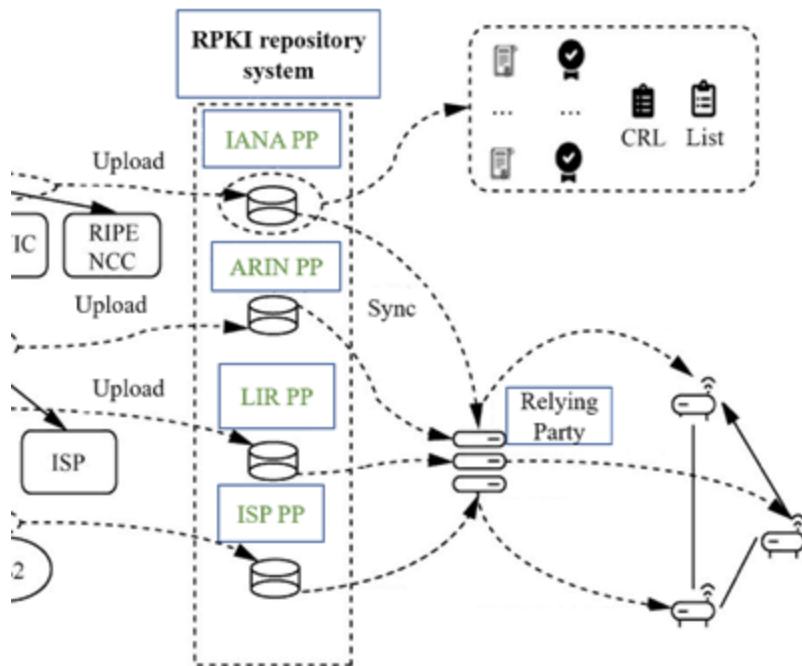


- RPKI Validators aka Relying Parties (RPs)
 - Download the RPKI repository from the RIRs
 - Validate the RPKI objects
 - Create a local “validated cache” with all valid ROAs
 - Validated ROA Payloads (VRPs) are tuples
 - > (ASN, ROA prefix, prefix length, max length)

https://www.juniper.net/documentation/en_US/release-independent/nce/information-products/pathway-pages/nce/nce-187-bgp-rpki-tn.pdf

Securing Routing Protocols (cont'd)

Route origin validation

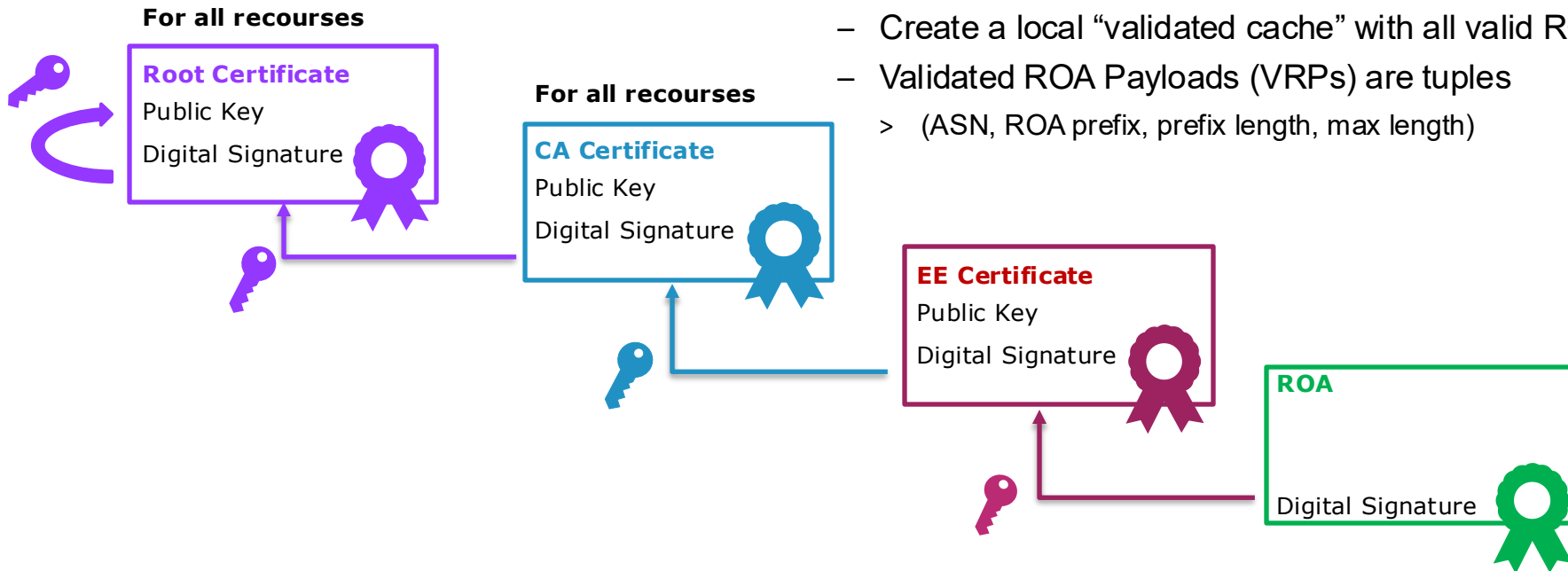


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Securing Routing Protocols (cont'd)

Route origin validation

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Securing Routing Protocols (cont'd)

Route origin validation

Router

- Get the VRPs through RP protocol
- Decide: accept or discard a new BGP announcement
 - BGP announcement covered if
 - > IP prefix and VRP IP prefix are identical for all bits specified by the VRP IP prefix length
 - BGP announcement a match if all hold
 - > VRP IP prefix covers the announcement
 - > VRP AS matches the announcement AS
 - > Length of the announcement prefix not greater than *the VRP MaxLength*
- Accept *valid* matching announcements
- Reject *invalid* announcements (IP prefix covered by a VRP but no match)
- Maybe accept *unknown ones with low preference (no VRP covers the announcement IP prefix)*



BGP Decision Process



Probing further on RPKI (cont'd)

Distributed RPKI

Issues

- RPKI trust anchors are usually RIRs
- Each end entity has great reliance on specific RIRs
 - No option to request certificates from different CAs/RIRs.
- Time consuming in cross-RIR manipulations at

Objectives/contributions:

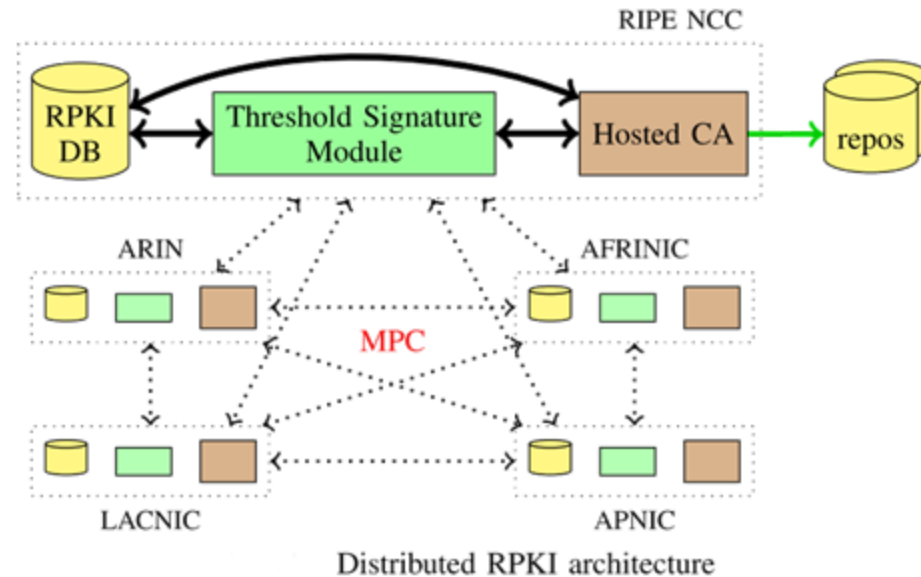
- Propose a distributed RPKI system across different RIRs
- Use a cooperative mechanism to sign the certificates.

K. Shrishak and H. Shulman, "Privacy Preserving and Resilient RPKI," IEEE INFOCOM 2021 - IEEE Conference on Computer Communications, 2021, pp. 1-10, doi: 10.1109/INFOCOM42981.2021.9488759.

Probing further on RPKI (cont'd)

Distributed RPKI

- A certificate is cooperatively signed by more than 1 RIR. (Utilizing threshold signature with ECDSA)
- The CAs can use the public key to verify the signature.





Probing further on RPKI (cont'd)

Distributed RPKI

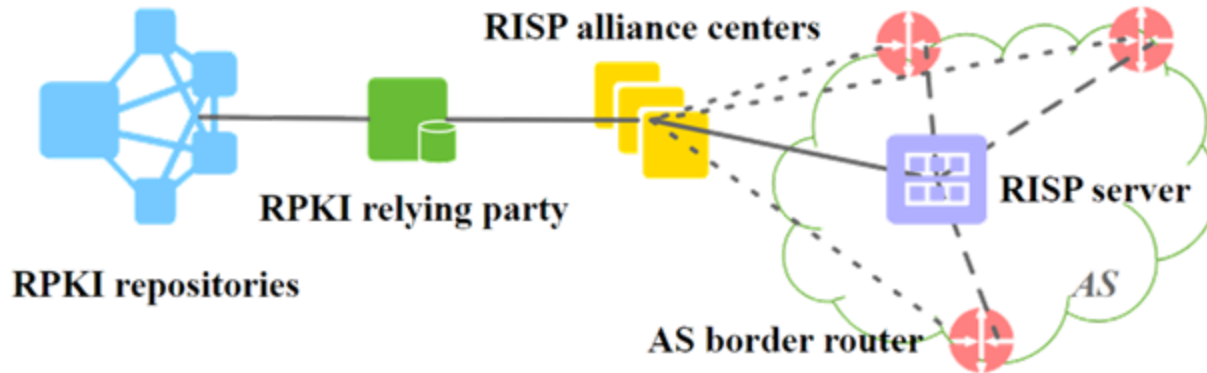
Key tech: threshold signatures (simplified explanation)

- Prerequisites
 - Each RIR holds a “security key share” (sk_i)
 - A public key to verify the certificate is known by CAs (generated by multiple sk_i)
 - The threshold number n is known (i.e., how many key shares are needed at least)
- Procedure
 - Two or more RIRs sign the certificate with the same content by their own sk_i
 - The hosted CA collect the certificate from more than one RIR
 - When the number of received certificates is equal to/larger than the threshold n , CAs start aggregating the certificates and verify the signature with the public key.

Probing further on RPKI (cont'd)

A RPKI-Based Inter-AS Source Protection mechanism

- The relying party is not directly connected to each border router
- A RSIP alliance center is deployed between the relying party and the border routers in ASes.



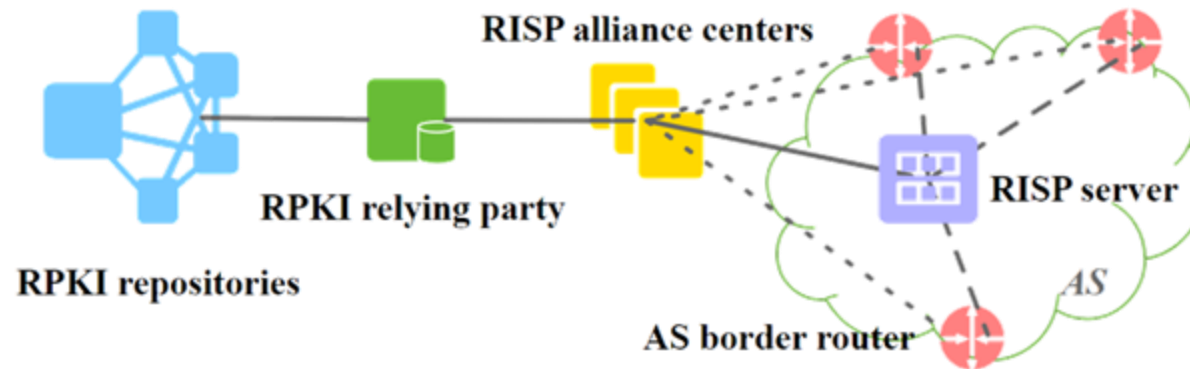
RISP architecture

Y. Jia, Y. Liu, G. Ren and L. He, "RISP: An RPKI-based inter-AS source protection mechanism," in Tsinghua Science and Technology, vol. 23, no. 1, pp. 1-12, Feb. 2018, doi: 10.26599/TST.2018.9010025.

Probing further on RPKI (cont'd)

A RPKI-Based Inter-AS Source Protection mechanism

- A RSIP alliance center connected with several RSIP servers
- A RSIP server is deployed as a “center controller” in each AS



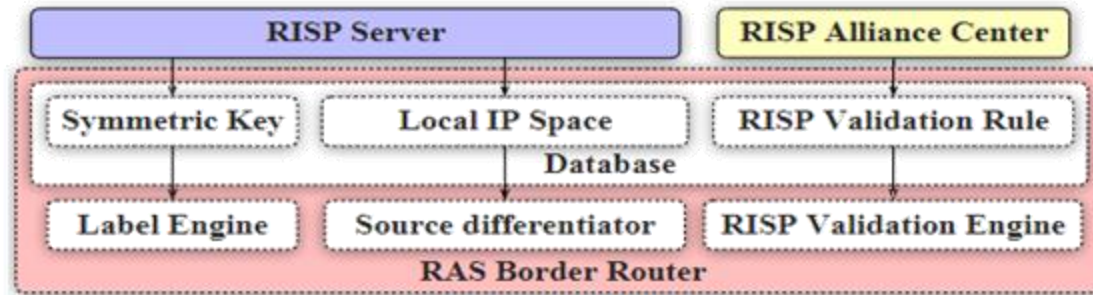
RSIP architecture

Probing further on RPKI (cont'd)

A RPKI-Based Inter-AS Source Protection mechanism

Labeling

- RISP server register a symmetric key to RSIP alliance center and install it in all border routers
- Border routers tag a label to the packets originated from this AS with the symmetric key.



RAS border router structure.



Probing further on RPKI (cont'd)

A RPKI-Based Inter-AS Source Protection mechanism

Filtering

- RISP alliance centers and RISP server push the rule of RISP validation to border routers (the validated IP prefix and their symmetric keys)
- Border routers will follow the rules to automatically filter the packets without the correct pair of ASN, IP prefix and symmetric key.

RISP Validation Rule		
IP	ASN	Symmetric Key
37.12.*.* /16	13 122	d42b2f1a3b12...
2001:2f35:* /48	370	1d43f1a3b25c...
...	...	...

RISP validation rule.



Probing further on RPKI (cont'd)

INR Reallocation and Reauthorization in RPKI

INR allocation conflicts

- A CA may accidentally or maliciously allocate the same set of IP prefixes to different customers or allocate INRs it does not own.
- Conflict types
 - A: The CA allocates/authorizes INRs that are not allocated by its superior CA to its customers/ROAs. This may result in conflicting INRs in two or more resource certificates whose issuers are different.
 - B: Two or more trust anchors allocate the same INRs to their customers. This results in conflicting INRs in resource certificates whose issuers are different.
 - C: The CA which is not a trust anchor allocates the same INRs to its customers. This results in conflicting INRs in resource certificates whose issuers are the same

H. Zou, D. Ma, Q. Shao and W. Mao, "The Horizontal INR Conflict-Detection Algorithm: Revealing INR Reallocation and Reauthorization in RPKI," 2021 IFIP/IEEE International Symposium on Integrated Network Management (IM), 2021, pp. 459-465.

INR: internet number resource



Probing further on RPKI (cont'd)

INR Reallocation and Reauthorization in RPKI

INR authorization conflicts

- For an AS0 ROA, prefixes described in this ROA, and any more specific prefixes, should not be used in inter-domain routing system.
- Conflict types
 - D: The CA allocates an IP prefix to its customer and also authorizes it to ASX ($X \neq 0$) for its own use. This results in conflicting INRs in a resource certificate and an ASX ROA ($X \neq 0$)
 - E: The CA reserves an IP prefix for future use but also authorizes it to ASX ($X \neq 0$) for its own use. This results in conflicting INRs in an AS0 ROA and an ASX ROA ($X \neq 0$)
 - F: The CA reserves an IP prefix for future use but also allocates it to its customer. This results in conflicting INRs in an AS0 ROA and a resource certificate



Probing further on RPKI (cont'd)

INR Reallocation and Reauthorization in RPKI

Solution: horizontal INR conflict detection algorithm

1. For a given IP prefix, the horizontal algorithm first finds the RC it belongs to
2. Then, it finds all RCs and ROAs issued by this RC and classifies these objects into three categories:
 - AS0 ROAs,
 - ASX ROAs($X \neq 0$)
 - RCs
3. If more than one RC exists (conflict C or B), we compare the authority key information of these RCs.
 - If same: conflict C
 - If not: conflict B

Note: Conflict A cannot be detected by this algorithm

RC: resource certificate



Probing further on RPKI (cont'd)

INR Reallocation and Reauthorization in RPKI

Solution: horizontal INR conflict detection algorithm (cont'd)

4. If there exists one AS0 ROA and ASX ROA($X \neq 0$) pair containing the same IP prefix.
It is conflict E
5. If an ASX ROA($X \neq 0$) and a RC or an AS0 ROA and a RC has the same IP prefix,
It is conflict D/F

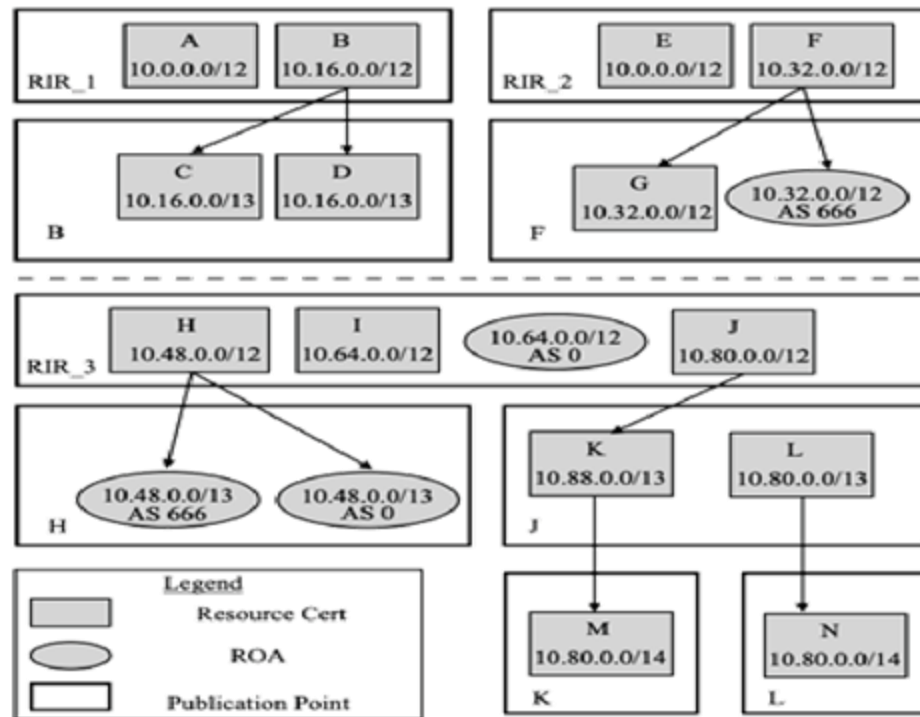
RC: resource certificate

Probing further on RPKI (cont'd)

INR Reallocation and Reauthorization in RPKI

Conflict summary and examples:

- Conflict C: RC C & D
- Conflict B: RC A & E
- Conflict D: RC G and ROA issued by RC F
- Conflict F: RC I and ROA issued by RIR_3
- Conflict E:
 - Exists one AS0 ROA and ASX ROA($X \neq 0$) pair containing the same IP prefix
 - ROAs issued by RC H



INR Conflict Model



Autumn 2025

Secure Routing

Ad hoc networks and BGP

Discussion leads
Marco and Ahmed

Panos Papadimitratos

www.eecs.kth.se/nss

